

Supplementary Material for Federated Adaptive Causal Estimation with High-Dimensional Covariates (FACE-HD)

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Supplementary Material for FACE-HD

This standalone Supplementary Material collects the asymptotic theory and additional numerical results supporting the main manuscript. The fold-level algorithm details are presented in the combined manuscript supplement. The sections below establish the asymptotic normality and variance estimation of each source-assisted cross-fitted estimator, report additional simulation results, and develop the aggregation theory underlying FACE-HD: the covariance structure, penalized weight estimation, source selection, and the oracle-limit guarantee that protects against negative transfer. The same notation as the main manuscript is used throughout; a detailed table of contents follows.

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We use the same mathematical notation as the main manuscript throughout this Supplemental Material. The theory is organized in a theorem-first format: setup and notation, assumptions, main limit statements, variance estimation, and then the supporting lemmas. Aggregation is treated in its own section, where the covariance structure, penalized weight estimation, source selection, and final oracle-limit theorem are stated before the proof details.

For readability, the proofs are grouped by the statistical conclusion they support. The pairwise theory proves the source-assisted cross-fitted expansion first: identification, leading influence-function CLT, high-dimensional nuisance rates, orthogonality, second-order remainders, and variance consistency. The aggregation theory then treats FACE-HD as an oracle-selection problem: shared-target covariance decomposition, uniform convergence of the quadratic variance objective, Wald-penalized source selection, asymptotic equivalence to oracle weights, and the final CLT/Slutsky argument.

S1 Asymptotic Normality of the Cross-Fitted Estimator

S1.1 Setup and Notation

This section is organized in the standard theorem-driven order used in high-dimensional inference. We first state the setup and assumptions, then give the main pairwise asymptotic normality result and variance estimator, then collect the supporting technical lemmas in the order they are used. The control-arm and TATE extension appears after the pairwise technical lemmas, because it is a direct reuse of the treated-arm argument.

We analyze the asymptotic properties of the K_f -fold cross-fitted estimator for a single target-source pair (t, s) , derived using the two-level cross-fitting strategy described in the

combined manuscript’s fold-level supplement. Let K_f be a fixed integer (e.g., 5 or 10). The data $\mathcal{D} = \mathcal{D}_t \cup \mathcal{D}_s$ is partitioned into K_f folds, $\mathcal{D} = \cup_{k=1}^{K_f} \mathcal{D}_k$, where $\mathcal{D}_k = \mathcal{D}_{t,k} \cup \mathcal{D}_{s,k}$. Let $N_k = |\mathcal{D}_k| \approx N/K_f$. For notational simplicity, we assume the folds are balanced within each site (e.g., N_t and N_s are divisible by K_f) so that $|\mathcal{D}_{t,k}| = N_t/K_f$ and $|\mathcal{D}_{s,k}| = N_s/K_f$ for all k ; the minor rounding effects in practice are $o_p(N^{-1/2})$ and do not affect the first-order results. For $r \in \{t, s\}$, $\mathbb{E}_r[\cdot]$ denotes expectation conditional on site membership $R = r$. Define the target-site potential outcome means $\mu^1 := \mathbb{E}_t[Y(1)]$ and $\mu^0 := \mathbb{E}_t[Y(0)]$, and the target average treatment effect (TATE) $\tau := \mu^1 - \mu^0$. For any function $f(D)$ of a single observation $D = (R, \mathbf{X}, A, Y)$, we write $\tilde{E}_k[f(D)] := \frac{1}{N_k} \sum_{i \in \mathcal{D}_k} f(D_i)$ for the empirical average over the combined k -th fold. Let $\mathcal{D}^{(-k)} = \mathcal{D} \setminus \mathcal{D}_k$ denote the out-of-fold data used for training nuisance models evaluated on fold k . The crucial aspect of cross-fitting is that the data used for training, $\mathcal{D}^{(-k)}$, is independent of the data used for evaluation, $\mathcal{D}_k$.

Let $\vartheta = (\boldsymbol{\alpha}^1, \boldsymbol{\gamma}_1)$ be the vector of nuisance parameters. For this fixed pair (t, s) we write $\boldsymbol{\alpha}^1 \equiv \boldsymbol{\alpha}_{t,s}^1$ and $\boldsymbol{\gamma}_1 \equiv \boldsymbol{\gamma}_{s,1}$. Let $\vartheta^* = (\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ denote the population targets of the calibrated estimation step (Eq. (6) and Eq. (7) in the main manuscript), and let $\hat{\vartheta}^{(-k)} = (\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)}, \hat{\boldsymbol{\gamma}}_{s,1}^{(-k)})$ be the nuisance estimators trained on data $\mathcal{D}^{(-k)}$ using the two-level cross-fitting procedure. Here the superscript $(-k)$ denotes the *final* two-level cross-fitted nuisance estimators used for evaluating fold k (including the fold-summed calibration over secondary folds), and hence $\hat{\vartheta}^{(-k)}$ is independent of $\mathcal{D}_k$. The inner construction follows the SMMAL two-level calibration device: for each outer fold k_1 , every secondary fold $k_2 \neq k_1$ supplies one calibrated-loss contribution, and the initial nuisance fits entering that contribution are trained on $\mathcal{D} \setminus (\mathcal{D}_{k_1} \cup \mathcal{D}_{k_2})$. The final $\hat{\vartheta}^{(-k_1)}$ is obtained by minimizing the average of these fold-specific calibrated losses, so it uses all outer-training folds while keeping the initial plug-in for each contribution disjoint from that contribution’s calibration fold.

To ensure numerical stability and theoretical regularity (bounded weights), we define a

truncation function $\mathcal{T}(x) = \text{sign}(x) \min(|x|, M_\tau)$, where $M_\tau = M_{\tau,N}$ is allowed to depend on N (notation suppressed). (We avoid the symbol $\tau(\cdot)$ to prevent collision with the TATE τ in the main manuscript.) Equivalently, one may define the working covariate vector $Z = \phi(\mathbf{X})$ and write all nuisance models directly in terms of Z . We keep the notation $\phi(\mathbf{X})$ to match the FACE-style basis notation and to distinguish raw covariates from the fixed feature/basis expansion. All tail, sparsity, and restricted-curvature conditions below are imposed on this working covariate vector. The truncated weighting function is $w_\tau(\mathbf{X}; \boldsymbol{\gamma}) := I(A = 1)/e^{\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\gamma})}$. The corresponding untruncated weight is $w(\mathbf{X}; \boldsymbol{\gamma}) := I(A = 1)/e^{\phi(\mathbf{X})^T \boldsymbol{\gamma}}$. The implementation permits separate fitting and inference radii, $M_{\tau,\text{fit}}$ and $M_{\tau,\text{inf}}$, and uses finite truncation both in the calibrated losses and in the final source correction, influence function, and variance estimator. Unless stated otherwise, both radii equal 5. Thus, in finite-sample estimator displays, an exponential source weight written as w is evaluated as w_τ with $M_\tau = M_{\tau,\text{inf}}$; retaining the shorter notation keeps the aggregation formulas readable.

The proofs use $w(\mathbf{X}; \boldsymbol{\gamma})$ as the untruncated limiting object. Under [Assumption S2\(c\)](#), the radii may increase with N so that truncation is asymptotically inactive at the population targets: in particular, the probability of clipping is $o(N^{-1})$, $w_\tau(\mathbf{X}; \boldsymbol{\gamma}_1^*) = w(\mathbf{X}; \boldsymbol{\gamma}_1^*)$ with sufficiently high probability, and $g'(\mathcal{T}(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})) = g'(\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*})$ likewise. The clipped and untruncated estimating equations then differ by $o_p(N^{-1/2})$, so their first-order limit theory agrees. If a fixed inference radius remains active with nonnegligible probability, the implementation instead defines a clipped estimating equation whose clipping-induced bias and variance must be assessed, for example through the stated M_τ sensitivity analysis (see also [Tan \(2020\)](#)).

The proofs below therefore use three conventions: foldwise IF components are centered when forming variance and covariance estimators; finite-sample clipping is represented by w_τ but

is asymptotically inactive under [Assumption S2\(c\)](#); and any finite-sample symmetrization or ridge stabilization used in the aggregation quadratic program is treated as a numerical device that vanishes asymptotically. Rather than relying on a standalone table, the Supplemental Material now states the role of each block in prose. [Lemma S1](#) identifies the target mean under either nuisance-model correctness; [Assumptions S1](#) to [S3](#) collect the sampling, overlap/moment, and nuisance-rate requirements; [Theorem S1](#) gives the pairwise cross-fitted CLT; [Lemma S2](#) proves variance consistency; and [Section S3](#) extends these ingredients to adaptive aggregation and source selection.

We assume $\phi(\mathbf{X})$ includes an intercept term. Consistent with the implementation and the convention in the main manuscript, every ℓ_1 penalty in a nuisance-fitting loss excludes that intercept; ordinary norms in assumptions, bounds, and rate statements include all coordinates unless stated otherwise. For completeness, the implementation guards against an exactly noncoercive empirical density-ratio loss in a sparse source arm. Write $\mathbf{m}$ for the target-side linear moment appearing in that loss. If feature j equals a constant c_j for every observation in the fitted source arm, then $\mathbf{d}_j = (-c_j, \mathbf{e}_j)$ lies in the null space of the source design with intercept, while $\mathbf{m}^T \mathbf{d}_j = m_j - c_j m_0$. Along $t\mathbf{d}_j$, the exponential term is unchanged and the non-intercept ℓ_1 penalty grows at slope λ_γ ; hence $\lambda_\gamma > |m_j - c_j m_0|$ is sufficient for coercivity in this coordinate-null direction. Before cross-validation, the implementation replaces every candidate by at least

$$\lambda_{\gamma, \text{support}} = \max_{j: \phi_j \text{ constant in the source arm}} |m_j - c_j m_0| + \epsilon_{\text{num}},$$

with a scale-adjusted strict numerical margin ϵ_{num} , and reports the selected penalty, floor, number of constant features, and whether the candidate path was constrained. This is a finite-sample optimization diagnostic, not evidence of identification: a persistent whole-site support failure violates the intended overlap regime and is screened rather than repaired by penalization. Under fixed-dimensional cell positivity, the activation probability vanishes

with the arm-specific training size, so the safeguard is asymptotically inactive and leaves the nuisance-rate assumptions below unchanged. For every nuisance cross-validation path, the implementation additionally requires a candidate penalty to satisfy the full-coordinate stopping rule and have finite validation loss in every inner fold. Partially valid candidates are not averaged over a smaller fold set: they receive infinite validation score and are excluded. The numbers of excluded fold fits and excluded penalties are retained as numerical diagnostics, and estimation stops if no candidate is valid across all folds. An optimizer-failed candidate is rolled back before the next pathwise warm start, so an unconverged coefficient vector cannot contaminate a later candidate. To avoid repeatedly exhausting the bounded iteration budget along a terminal low-penalty tail, the implementation stops a fold-specific descending path after three consecutive nonconverged candidates and leaves the remaining candidates at their preallocated infinite score. Isolated failures therefore retain two opportunities to recover. The skipped-fold count is reported separately, and the locked $p = 100$ audit described in the main manuscript requires exact agreement of every scientific and final-fit field, including the full cutoff and inference-radius sensitivity sidecar; only the mechanically affected invalid-candidate counts may differ, and each difference must be reconciled to the reported skip count. Let $p := \dim(\phi(\mathbf{X}))$. When the feature expansion is fixed in advance, one may equivalently regard $\phi(\mathbf{X})$ as the working covariate vector. For a real-valued random variable Z , $\|Z\|_{\psi_2}$ denotes the sub-Gaussian Orlicz norm. For a vector $v \in \mathbb{R}^p$, $\|v\|_0$ denotes its sparsity level. For canonical exponential-family GLMs, we use the standard notation that the negative log-likelihood takes the form $-Y\theta + \Psi(\theta)$ with linear predictor $\theta = \phi(\mathbf{X})^T \boldsymbol{\alpha}$, and that $g(\theta) = \Psi'(\theta)$; $g'(\theta)$ denotes the derivative with respect to θ . We also use the shorthand $g(\phi(\mathbf{X}); \boldsymbol{\alpha}) \equiv g(\phi(\mathbf{X})^T \boldsymbol{\alpha})$. To avoid confusion, $\phi(\mathbf{X})$ always denotes the feature/basis expansion, $g(\cdot)$ denotes the GLM response function, and the Orlicz norm $\|\cdot\|_{\psi_2}$ is unrelated to the feature map.

As in the main manuscript, the target estimand is the target-site treated potential outcome mean $\mu^1 \equiv \mathbb{E}_t[Y(1)]$. Define the target-site treated regression function by $m_{t,1}(\mathbf{X}) := \mathbb{E}_t[Y \mid A = 1, \mathbf{X}]$. The K_f -fold cross-fitted estimator for μ^1 is:

$$\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \left\{ \tilde{E}_{t,k} \left[g(\phi(\mathbf{X}); \hat{\alpha}_{t,s}^{1,(-k)}) \right] + \tilde{E}_{s,k} \left[w(\mathbf{X}; \hat{\gamma}_{s,1}^{(-k)}) \left(Y - g(\phi(\mathbf{X}); \hat{\alpha}_{t,s}^{1,(-k)}) \right) \right] \right\}.$$

Here,

$$\tilde{E}_{t,k}[q(D)] := \frac{1}{N_{t,k}} \sum_{i \in \mathcal{D}_{t,k}} q(D_i), \quad \tilde{E}_{s,k}[q(D)] := \frac{1}{N_{s,k}} \sum_{i \in \mathcal{D}_{s,k}} q(D_i),$$

denote sample averages over the k -th fold of the target and source data, with $N_{t,k} = |\mathcal{D}_{t,k}|$, $N_{s,k} = |\mathcal{D}_{s,k}|$, and $N_k = N_{t,k} + N_{s,k}$. Let $N = N_t + N_s$ be the total sample size, and let $\rho_t = N_t/N$ and $\rho_s = N_s/N$. We write $\mathbb{E}[\cdot]$ for expectation under the pooled distribution, i.e., $\mathbb{E}[q(D)] = \rho_t \mathbb{E}_t[q(D)] + \rho_s \mathbb{E}_s[q(D)]$. Throughout the asymptotics we consider the standard two-sample (multi-sample) regime with sampling fractions bounded away from 0 and 1, i.e., $N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s \in (0, 1)$. Under overlap/positivity for the relevant treatment arm, the effective sample sizes within each arm also scale linearly with N (e.g., $\sum_{i \in \mathcal{D}_s} I(A_i = 1) = \Theta_p(N_s)$), so writing high-dimensional rates in terms of N is equivalent (up to constants) to writing them in terms of the arm-specific sample sizes. Under the balanced-fold assumption above, the estimator can equivalently be written using the combined-fold empirical average as

$$\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k \left[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) \right],$$

which matches the $\tilde{E}_k$ notation used in the remainder decomposition below.

Define the (uncentered) score contribution at a generic nuisance parameter $\vartheta = (\alpha^1, \gamma_1)$:

$$\tilde{\varphi}_{t,s}(D_i; \vartheta) := \frac{I(R_i = t)}{\rho_t} g(\phi(\mathbf{X}_i); \alpha^1) + \frac{I(R_i = s)}{\rho_s} w(\mathbf{X}_i; \gamma_1) \{Y_i - g(\phi(\mathbf{X}_i); \alpha^1)\}.$$

Define the corresponding population estimand at ϑ as

$$\mu^1(\vartheta) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta)] = \mathbb{E}_t[g(\phi(\mathbf{X}); \alpha^1)] + \mathbb{E}_s[w(\mathbf{X}; \gamma_1) \{Y - g(\phi(\mathbf{X}); \alpha^1)\}].$$

Lemma S1 (Identification / double robustness for $\mu^1(\vartheta^*)$). *Assume the target causal parameter is identified as $\mu^1 = \mathbb{E}_t[Y(1)]$ (e.g., by consistency and ignorability/overlap on the target site), so that $\mu^1 = \mathbb{E}_t[m_{t,1}(\mathbf{X})]$. Recall $\phi(\mathbf{X})$ includes an intercept. Let $\vartheta^* = (\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ denote the population targets of the calibrated losses. Then $\mu^1(\vartheta^*) = \mu^1$ holds under either of the following conditions.*

1. **Outcome-model correctness.** *If $g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*}) = m_{t,1}(\mathbf{X})$ almost surely, then the calibrated outcome loss implies the population moment condition $\mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \phi(\mathbf{X}) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}] = \mathbf{0}$ (see [Lemma S5\(ii\)](#) below). In particular, the intercept component yields $\mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}] = 0$, and therefore $\mu^1(\vartheta^*) = \mathbb{E}_t[g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})] = \mathbb{E}_t[m_{t,1}(\mathbf{X})] = \mu^1$.*
2. **Weight-model correctness.** *Suppose the weighting model is correct in the sense that for any square-integrable measurable function $h(\mathbf{X})$,*

$$\mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) h(\mathbf{X})] = \mathbb{E}_t[h(\mathbf{X})],$$

and, for the treated potential outcome, the usual covariate-shift/transportability condition holds for this pair (t, s) : $\mathbb{E}_s[Y(1) \mid \mathbf{X}] = \mathbb{E}_t[Y(1) \mid \mathbf{X}]$ almost surely. Then for any outcome model $m(\mathbf{X})$,

$$\mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \{Y(1) - m(\mathbf{X})\}] = \mathbb{E}_t[Y(1) - m(\mathbf{X})],$$

and hence $\mu^1(\boldsymbol{\alpha}, \boldsymbol{\gamma}_1^) = \mu^1$ for any $\boldsymbol{\alpha}$, in particular $\mu^1(\vartheta^*) = \mu^1$ even if the outcome model is misspecified.*

Define the centered influence function at the population parameters ϑ^* by centering each site's contribution at its own site mean:

$$\begin{aligned} \varphi_{t,s}(D_i; \vartheta^*) &= \frac{I(R_i = t)}{\rho_t} \left(g(\phi(\mathbf{X}_i); \boldsymbol{\alpha}^{1,*}) - \mu_t^1(\boldsymbol{\alpha}^{1,*}) \right) \\ &\quad + \frac{I(R_i = s)}{\rho_s} \left(w(\mathbf{X}_i; \boldsymbol{\gamma}_1^*) \{Y_i - g(\phi(\mathbf{X}_i); \boldsymbol{\alpha}^{1,*})\} - \delta_s^1(\vartheta^*) \right), \end{aligned}$$

where $\mu_t^1(\boldsymbol{\alpha}^{1,*}) = \mathbb{E}_t[g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})]$ and $\delta_s^1(\vartheta^*) = \mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*)\{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}]$, so that $\mathbb{E}_t[\varphi_{t,s}] = \mathbb{E}_s[\varphi_{t,s}] = 0$. Its full-sample average reproduces the (uncentered) estimation error, $\frac{1}{N} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) = \tilde{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)] - \mu^1(\vartheta^*)$, because the site-specific centering constants cancel in the pooled sum. However, the within-site-centered form $\varphi_{t,s}$ must be distinguished from the pooled deviation $\tilde{\varphi}_{t,s}(D; \vartheta^*) - \mu^1(\vartheta^*)$: the two share the same mean but have different per-observation variances, because the ρ_t^{-1}, ρ_s^{-1} scaling makes the site-conditional means of $\tilde{\varphi}_{t,s}$ differ ($\mathbb{E}_t[\tilde{\varphi}_{t,s}] = \mu_t^1/\rho_t$, $\mathbb{E}_s[\tilde{\varphi}_{t,s}] = \delta_s^1/\rho_s$). More generally, for any ϑ we write $\varphi_{t,s}(D; \vartheta)$ for the same within-site-centered contribution with (μ_t^1, δ_s^1) evaluated at ϑ . The asymptotic variance is governed by the within-site-centered form:

$$\begin{aligned} V_{t,s}^* := \text{Var}(\varphi_{t,s}(D; \vartheta^*)) &= \frac{1}{\rho_t} \text{Var}_t(g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})) \\ &\quad + \frac{1}{\rho_s} \text{Var}_s(w(\mathbf{X}; \boldsymbol{\gamma}_1^*)\{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}). \end{aligned}$$

which differs from the pooled variance $\text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta^*)) = V_{t,s}^* + \{\rho_t^{-1}(\mu_t^1)^2 + \rho_s^{-1}(\delta_s^1)^2 - (\mu^1)^2\}$ by a non-vanishing between-site term.

Definition S1 (Sparsity bookkeeping). Let

$$s_\alpha := \|\boldsymbol{\alpha}^{1,*}\|_0, \quad s_\gamma := \|\boldsymbol{\gamma}_1^*\|_0.$$

When the initial nuisance estimators have distinct population targets, define $s_{\alpha,\text{init}} := \|\boldsymbol{\alpha}_{\text{init}}^{1,*}\|_0$ and $s_{\gamma,\text{init}} := \|\boldsymbol{\gamma}_{1,\text{init}}^*\|_0$. These are bookkeeping quantities for the high-dimensional rates rather than separate modeling assumptions. The growth restrictions are imposed explicitly in the theorems and, when initial nuisance rates are needed, we assume $s_{\alpha,\text{init}} + s_{\gamma,\text{init}} = O(s_\alpha + s_\gamma)$.

S1.2 Assumptions

The assumptions below follow the compact itemized style used in FACE and SMMAL: each assumption starts with fixed constants and then lists the required pieces as labeled subconditions. [Assumption S1](#) collects working-feature and sampling regularity; [Assumption S2](#)

collects positivity, moments, smoothness, and truncation regularity; and [Assumption S3](#) collects the nuisance-estimation conditions. Sparsity levels are defined in [Definition S1](#), and their allowed growth is stated directly in the theorems.

Assumption S1 (Working-Covariate and Sampling Regularity). For a fixed constant $M < \infty$, assume:

- (a) (*Working-feature tails*) The working covariates $\phi(\mathbf{X})$ include an intercept and satisfy

$$\max_{1 \leq j \leq p} \|\phi_j(\mathbf{X})\|_{\psi_2} \leq M$$

under both $R = t$ and $R = s$.

- (b) (*Independent two-site sampling*) The target and source samples are independent, observations are i.i.d. within each site, and the site fractions satisfy $N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s \in (0, 1)$.

Assumption S2 (Overlap, Smoothness, and Moments). For fixed constants $M < \infty$ and $\delta > 0$, assume:

- (a) (*Overlap and bounded local weights*) The ideal density-ratio weight is well defined on the relevant target support: $P_s\{A = 1 \mid \mathbf{X}\} > 0$ almost surely on that support. In a neighborhood $\mathcal{N}(\gamma_1^*)$ of the population tilting target,

$$\sup_{\gamma \in \mathcal{N}(\gamma_1^*)} \mathbb{E}_s[w(\mathbf{X}; \gamma)^{2+\delta}] \leq M,$$

and, coordinatewise,

$$\max_{1 \leq j \leq p} \sup_{\gamma \in \mathcal{N}(\gamma_1^*)} \mathbb{E}_s[w(\mathbf{X}; \gamma)^{2+\delta} |\phi_j(\mathbf{X})|^{2+\delta}] \leq M.$$

A simple sufficient condition is an upper-overlap bound $w(\mathbf{X}; \gamma) \leq M$ almost surely on $\mathcal{N}(\gamma_1^*)$, together with coordinatewise $(2 + \delta)$ moments of $\phi(\mathbf{X})$. No uniform bound on $\|\phi(\mathbf{X})\|_2$ is imposed; in high dimensions this norm may scale as $\sqrt{p}$.

- (b) (*Smoothness and bounded moments*) The GLM mean function g is continuously differentiable with

$$\sup_{u \in \mathbb{R}} |g'(u)| \leq M, \quad |g'(u_1) - g'(u_2)| \leq M|u_1 - u_2|.$$

Moreover, under both $R = t$ and $R = s$,

$$\mathbb{E}[|Y|^{2+\delta} \mid R = r] \leq M, \quad \sup_{\vartheta \in \mathcal{N}(\vartheta^*)} \mathbb{E}[|\tilde{\varphi}_{t,s}(D; \vartheta)|^{2+\delta}] \leq M.$$

Products of any two influence-function components used in the aggregation covariance estimators are uniformly integrable. The last display is the uniform integrability condition used for the influence-function expansion and variance consistency. It follows, for example, from the weight envelope in part (a), sub-Gaussian working features, and sub-Gaussian or bounded treated outcomes.

- (c) (*Asymptotically inactive truncation*) With $M_{\tau,N}$ as in the definition of $\mathcal{T}(\cdot)$,

$$P\left(|\phi(\mathbf{X})^T \boldsymbol{\alpha}^{1,*}| > M_{\tau,N} \mid R = r\right) = o(N^{-1}) \quad \text{for } r \in \{t, s\},$$

$$P\left(|\phi(\mathbf{X})^T \boldsymbol{\gamma}_1^*| > M_{\tau,N} \mid R = s\right) = o(N^{-1}).$$

This condition guarantees that finite-sample clipping in the calibrated losses and final source correction has no first-order effect on the limiting estimating equation.

Assumption S3 (Nuisance Model Properties). For fixed constants $M < \infty$ and $\kappa > 0$, assume:

- (a) (*Restricted curvature*) The population versions of the calibrated losses in [Eq. \(6\)](#) and [Eq. \(7\)](#) are convex and satisfy restricted strong convexity with curvature at least κ in neighborhoods of ϑ^* compatible with the relevant sparse cones.
- (b) (*Cross-fitting and bounded fitted scores*) For each fold k , the nuisance estimator $\hat{\vartheta}^{(-k)}$ is trained on $\mathcal{D}^{(-k)}$ only and is independent of the evaluation fold $\mathcal{D}_k$. With probability tending to one, the fitted nuisance functions used on the evaluation folds lie in the local neighborhoods where the bounds in [Assumption S2\(a\)–\(b\)](#) hold.

- (c) (*Two-level alignment*) Let $(\boldsymbol{\alpha}_{\text{init}}^{1,*}, \boldsymbol{\gamma}_{1,\text{init}}^*)$ denote the population limits of the initial nuisance estimators used inside the calibrated losses. These limits coincide with the calibrated targets:

$$\boldsymbol{\alpha}_{\text{init}}^{1,*} = \boldsymbol{\alpha}^{1,*}, \quad \boldsymbol{\gamma}_{1,\text{init}}^* = \boldsymbol{\gamma}_1^*.$$

- (d) (*Rate of estimation*) Uniformly over the fixed number of outer folds,

$$\|\hat{\boldsymbol{\alpha}}_{t,s}^{1,(-k)} - \boldsymbol{\alpha}^{1,*}\|_2 + \|\hat{\boldsymbol{\gamma}}_{s,1}^{(-k)} - \boldsymbol{\gamma}_1^*\|_2 = O_p\left(\sqrt{\frac{(s_\alpha + s_\gamma) \log(p)}{N}}\right),$$

with analogous ℓ_1 rates of order $O_p\{(s_\alpha + s_\gamma)\sqrt{\log(p)/N}\}$. Here $\|\cdot\|_2$ and $\|\cdot\|_1$ denote coefficient-vector norms. These rates are the high-dimensional nuisance input to the asymptotic expansion. They follow from [Lemma S4](#) under the stated restricted-curvature and score-stability conditions; the sharpened foldwise score-centering condition used in [Hou et al. \(2025\)](#) is one sufficient route, while the general version is also admissible provided the propagated plug-in error is of the same order.

Remark S1 (On the alignment condition). [Assumption S3\(c\)](#) is a technical device to avoid tracking multiple population limits induced by the nested/coupled optimization. It is satisfied, for example, when (i) the population calibrated losses have unique minimizers $(\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ (by strict convexity in the relevant restricted set), and (ii) the mapping from an initial plug-in value to the population minimizer of the next calibrated update is continuous at $(\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$, so that any consistent initialization converges to the same fixed point. In practice, the same calibrated estimating equations are solved throughout the nuisance-estimation pipeline, making the alignment natural under uniqueness.

S1.3 Main Pairwise Result and Variance Estimation

Theorem S1 (Asymptotic Normality of $\hat{\mu}_{t,s}^{1,\text{CF}}$). *Suppose [Assumptions S1](#) to [S3](#) hold. Assume ϑ^* is the unique population target pair induced by the calibrated losses under the*

alignment condition in [Assumption S3\(c\)](#), implying [Lemma S5](#) holds. If further, at least one nuisance model (outcome or weighting) is correctly specified in the sense of [Lemma S1](#) (so $\mu^1(\vartheta^*) = \mu^1$), the calibrated nuisance sparsities satisfy

$$\frac{(s_\alpha + s_\gamma)^2 \log^2(p)}{N} \longrightarrow 0,$$

and $V_{t,s}^* > 0$, then

$$\sqrt{N} \left(\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

Here s_α and s_γ are the calibrated population sparsities from [Definition S1](#). If the initial nuisance estimators are estimated by separate penalized fits, their population sparsities are $s_{\alpha,\text{init}}$ and $s_{\gamma,\text{init}}$ and are assumed to be of the same order. Their stochastic effect enters only through the alignment and rate conditions in [Assumption S3\(c\)–\(d\)](#). The finite-sample KKT perturbation from the ℓ_1 penalties is likewise absorbed in these nuisance-rate conditions and is $o_p(N^{-1/2})$ under the stated sparsity scaling.

Proof. By [Lemma S1](#), the stated nuisance-correctness condition implies $\mu^1(\vartheta^*) = \mu^1$, so all expansions below are centered at the target estimand.

$$S_N := \frac{1}{N} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*), \quad \text{Rem}_N := \hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 - S_N.$$

Step 1 (Limit of the leading term). The summands in S_N are independent across the target and source samples, have mean zero, and have uniformly bounded moments of order $2 + \delta$ by [Assumptions S1](#) and [S2](#). Hence [Lemma S3](#) gives

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*),$$

so $\sqrt{N}S_N$ has the desired normal limit.

Step 2 (Foldwise decomposition of the remainder). Using the foldwise representation in [Section S1.1](#),

$$\hat{\mu}_{t,s}^{1,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k \left[\tilde{\varphi}_{t,s} \{D; \hat{\vartheta}^{(-k)}\} \right].$$

Because the folds are balanced up to negligible rounding error,

$$S_N = \frac{1}{K_f} \sum_{k=1}^{K_f} \tilde{E}_k \left[\tilde{\varphi}_{t,s}(D; \vartheta^*) - \mu^1 \right].$$

Since $\mu^1 = \mathbb{E}\{\tilde{\varphi}_{t,s}(D; \vartheta^*)\}$ by the identification step, we may add and subtract both $\tilde{\varphi}_{t,s}(D; \vartheta^*)$ and the conditional mean of the fold-specific plug-in difference:

$$\text{Rem}_N = \frac{1}{K_f} \sum_{k=1}^{K_f} (R_{1,k} + R_{2,k}),$$

where

$$R_{1,k} = \left(\tilde{E}_k - \mathbb{E}[\cdot \mid \mathcal{D}^{(-k)}] \right) \left[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right]$$

and

$$R_{2,k} = \mathbb{E}[\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \mid \mathcal{D}^{(-k)}].$$

Step 3 (Conditional empirical-process term). Let $\Delta_k(D) = \tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*)$.

Conditional on $\mathcal{D}^{(-k)}$, cross-fitting makes the evaluation-fold summands independent and mean zero after subtracting the conditional expectation. Therefore

$$\text{Var}(R_{1,k} \mid \mathcal{D}^{(-k)}) \leq \frac{1}{N_k} \mathbb{E}[\Delta_k(D)^2 \mid \mathcal{D}^{(-k)}].$$

By the local L_2 stability in [Lemma S6](#) (Part 2),

$$\mathbb{E} \left[\left(\tilde{\varphi}_{t,s}(D; \hat{\vartheta}^{(-k)}) - \tilde{\varphi}_{t,s}(D; \vartheta^*) \right)^2 \mid \mathcal{D}^{(-k)} \right] \leq C \left(\|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_2^2 + \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_2^2 \right),$$

and [Lemma S4](#) yields, uniformly over the fixed number of folds,

$$|R_{1,k}| = O_p \left(\frac{\|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_2 + \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_2}{\sqrt{N}} \right) = O_p \left(\frac{\sqrt{(s_\alpha + s_\gamma) \log(p)}}{N} \right) = o_p(N^{-1/2}).$$

Step 4 (Orthogonal bias term). By [Lemma S6](#) (Part 1), whose proof uses the calibrated moment equations in [Lemma S5](#), the conditional bias remainder is second order:

$$|R_{2,k}| \leq C \cdot \left(\|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_2^2 + \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_2^2 + \|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_2 \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_2 \right)$$

and hence, by [Lemma S4](#),

$$|R_{2,k}| = O_p\left(\frac{s_\alpha \log(p)}{N}\right) + O_p\left(\frac{s_\gamma \log(p)}{N}\right) + O_p\left(\frac{\sqrt{s_\alpha s_\gamma} \log(p)}{N}\right) = O_p\left(\frac{(s_\alpha + s_\gamma) \log(p)}{N}\right).$$

Because K_f is fixed,

$$|\text{Rem}_N| = O_p\left(\frac{\sqrt{(s_\alpha + s_\gamma) \log(p)}}{N}\right) + O_p\left(\frac{(s_\alpha + s_\gamma) \log(p)}{N}\right).$$

The sparsity condition $(s_\alpha + s_\gamma)^2 \log^2(p)/N \rightarrow 0$ implies $\sqrt{N} \text{Rem}_N = o_p(1)$. Therefore

$$\sqrt{N} (\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) + o_p(1).$$

Together with [Lemma S3](#), this yields $\sqrt{N} (\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*)$ by Slutsky's theorem. $\square$

Lemma S2 (Variance Estimation). *The asymptotic variance $V_{t,s}^*$ is consistently estimated by the within-site-centered plug-in built from the cross-fitted, out-of-fold nuisance estimates $\hat{g}^{(-k)}$:*

$$\hat{V}_{t,s} = \frac{1}{\rho_t} \widehat{\text{Var}}_t(g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}^{(-k(i))})) + \frac{1}{\rho_s} \widehat{\text{Var}}_s(w(\mathbf{X}; \hat{\boldsymbol{\gamma}}^{(-k(i))})\{Y - g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}^{(-k(i))})\}) \xrightarrow{p} V_{t,s}^*,$$

where $\widehat{\text{Var}}_r(U) = \tilde{E}_r[U^2] - (\tilde{E}_r[U])^2$ is the sample variance over site $r \in \{t, s\}$, each observation evaluated at its out-of-fold nuisance estimate $\hat{\vartheta}^{(-k(i))}$. Centering is performed within each site (at $\tilde{E}_t$ and $\tilde{E}_s$ separately), not at the pooled mean $\hat{\mu}_{t,s}^{1,\text{CF}}$; the reason is given at the end of the proof. For consistency of the variance estimator itself no lower bound is needed beyond finiteness. For Wald inference and studentization we assume, as in [Theorem S1](#), that $V_{t,s}^*$ is bounded away from zero.

Proof. Write $\hat{V}_{t,s} = \rho_t^{-1} \widehat{\text{Var}}_t(\hat{A}) + \rho_s^{-1} \widehat{\text{Var}}_s(\hat{B})$ with the cross-fitted site integrands

$$\hat{A}_i = g(\phi(\mathbf{X}_i); \hat{\boldsymbol{\alpha}}^{(-k(i))}), \quad \hat{B}_i = w(\mathbf{X}_i; \hat{\boldsymbol{\gamma}}^{(-k(i))})\{Y_i - g(\phi(\mathbf{X}_i); \hat{\boldsymbol{\alpha}}^{(-k(i))})\}.$$

The consistency of each within-site sample variance relies on three ingredients:

1. *Nuisance Consistency:* By [Lemma S4](#), $\|\hat{\vartheta}^{(-k)} - \vartheta^*\|_2 = o_p(1)$.

2. *L_2 -Lipschitz Continuity*: Under [Assumption S2\(a\)–\(b\)](#), the site integrands are locally Lipschitz in ϑ in mean-square: there exists $C < \infty$ such that for $U \in \{A, B\}$, $r \in \{t, s\}$, and ϑ_1, ϑ_2 in a neighborhood of ϑ^* ,

$$\mathbb{E}_r \left[(U(D; \vartheta_1) - U(D; \vartheta_2))^2 \right] \leq C \|\vartheta_1 - \vartheta_2\|_2^2,$$

by the bounded-slope/Lipschitz property of g and the weight-moment conditions, exactly as in [Lemma S6\(2\)](#).

3. *Uniform Integrability*: By the $(2 + \delta)$ moment condition in [Assumption S2\(b\)](#), $\{\hat{A}^2\}_{k,N}$ and $\{\hat{B}^2\}_{k,N}$ are uniformly integrable.

Step 1 (Conditional LLN within a site). Fix a site $r \in \{t, s\}$, a fold k , and let $U \in \{A, B\}$ be the corresponding integrand. Conditional on the training sample $\mathcal{D}^{(-k)}$, the nuisance estimate $\hat{\vartheta}^{(-k)}$ is fixed and independent of $\mathcal{D}_{r,k}$, so the summands $\{U(D_i; \hat{\vartheta}^{(-k)}) : i \in \mathcal{D}_{r,k}\}$ are i.i.d. with finite $(2 + \delta)$ moments uniformly in N by Ingredient 3. By the conditional weak law of large numbers,

$$\begin{aligned} \tilde{E}_{r,k} [U(\hat{\vartheta}^{(-k)})] - \mathbb{E}_r [U(\hat{\vartheta}^{(-k)}) \mid \mathcal{D}^{(-k)}] &= o_p(1), \\ \tilde{E}_{r,k} [U(\hat{\vartheta}^{(-k)})^2] - \mathbb{E}_r [U(\hat{\vartheta}^{(-k)})^2 \mid \mathcal{D}^{(-k)}] &= o_p(1). \end{aligned}$$

Since K_f is fixed, averaging over the evaluation folds of site r gives the same statements with $\tilde{E}_{r,k}$ replaced by the pooled within-site average $\tilde{E}_r$.

Step 2 (Plug-in stability). By nuisance consistency (Ingredient 1) and the L_2 -Lipschitz property (Ingredient 2), conditional Jensen gives

$$\left| \mathbb{E}_r [U(\hat{\vartheta}^{(-k)}) \mid \mathcal{D}^{(-k)}] - \mathbb{E}_r [U(\vartheta^*)] \right| \leq \left(\mathbb{E}_r [(U(\hat{\vartheta}^{(-k)}) - U(\vartheta^*))^2 \mid \mathcal{D}^{(-k)}] \right)^{1/2} \rightarrow_p 0,$$

and, using $(a^2 - b^2) = (a - b)(a + b)$ with conditional Cauchy–Schwarz and the $(2 + \delta)$ bound (Ingredient 3),

$$\mathbb{E}_r [U(\hat{\vartheta}^{(-k)})^2 \mid \mathcal{D}^{(-k)}] \rightarrow_p \mathbb{E}_r [U(\vartheta^*)^2].$$

Combining with Step 1,

$$\widehat{\text{Var}}_r(\widehat{U}) = \widetilde{E}_r[\widehat{U}^2] - (\widetilde{E}_r[\widehat{U}])^2 \rightarrow_p \mathbb{E}_r[U(\vartheta^*)^2] - (\mathbb{E}_r[U(\vartheta^*)])^2 = \text{Var}_r(U(\vartheta^*)).$$

Step 3 (Combine sites). Applying Step 2 with $U = A$ on $r = t$ and $U = B$ on $r = s$, and using that the site fractions ρ_t, ρ_s are fixed and bounded away from 0,

$$\begin{aligned} \widehat{V}_{t,s} &= \rho_t^{-1} \widehat{\text{Var}}_t(\widehat{A}) + \rho_s^{-1} \widehat{\text{Var}}_s(\widehat{B}) \\ &\rightarrow_p \frac{1}{\rho_t} \text{Var}_t(g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})) \\ &\quad + \frac{1}{\rho_s} \text{Var}_s(w(\mathbf{X}; \boldsymbol{\gamma}_1^*)\{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}) = V_{t,s}^*. \end{aligned}$$

the last equality by the variance decomposition recorded with the definition of $V_{t,s}^*$.

Why within-site centering is necessary. Centering instead at the pooled mean, i.e. using $\frac{1}{K_f} \sum_k \widetilde{E}_k[(\widetilde{\varphi}_{t,s}(D; \widehat{\vartheta}^{(-k)}) - \widehat{\mu}_{t,s}^{1,\text{CF}})^2]$, would converge to the pooled variance

$$\text{Var}(\widetilde{\varphi}_{t,s}(D; \vartheta^*)) = V_{t,s}^* + \left\{ \rho_t^{-1} (\mu_t^1)^2 + \rho_s^{-1} (\delta_s^1)^2 - (\mu^1)^2 \right\},$$

overestimating $V_{t,s}^*$ by the non-vanishing between-site term. This term arises precisely because the ρ_r^{-1} -scaled contribution $\widetilde{\varphi}_{t,s}$ has unequal site-conditional means ($\mathbb{E}_t[\widetilde{\varphi}_{t,s}] = \mu_t^1/\rho_t$ versus $\mathbb{E}_s[\widetilde{\varphi}_{t,s}] = \delta_s^1/\rho_s$); subtracting each site's own mean removes it. This is the pairwise analogue of the within-group (WSS-only) centering used for the aggregated estimator in [Remark S2](#). □

Remark S2 (Unified cross-fitted variance estimator for aggregation). This remark records the foldwise direct-TATE implementation. For arm $a \in \{0, 1\}$, let $\widehat{\Phi}_i^a(\eta)$ be the uncentered site-scaled pseudo-value obtained from the display below with $I(A = a)$, the corresponding arm nuisance estimates, and the *same* weight vector η .

Unified estimating equation. For each observation i evaluated on fold $\mathcal{D}_{k(i)}$, define the

uncentered IF

$$\hat{\Phi}_i^a(\eta) := \begin{cases} \frac{N_{\text{all}}}{N_t} \left[(1 - \sum_j \eta_j) \tilde{Y}_{ot,i}^a + \sum_j \eta_j g(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^a) \right], & i \in \mathcal{D}_t, \\ \frac{N_{\text{all}}}{N_{s_j}} \eta_j w(\mathbf{X}_i; \hat{\gamma}_{s_j,a}) \{Y_i - g(\phi(\mathbf{X}_i); \hat{\alpha}_{t,s_j}^a)\}, & i \in \mathcal{D}_{s_j}, \end{cases}$$

where all nuisance functions are evaluated out of fold and

$$\tilde{Y}_{ot,i}^a = \frac{I(A_i = a)}{\hat{\pi}_{ot,a}(\mathbf{X}_i)} \{Y_i - \hat{g}_{ot}^a(\mathbf{X}_i)\} + \hat{g}_{ot}^a(\mathbf{X}_i).$$

The direct-TATE pseudo-value is formed before variance calculation: $\hat{\Phi}_i^\tau = \hat{\Phi}_i^1(\hat{\eta}^{(-k(i))}) - \hat{\Phi}_i^0(\hat{\eta}^{(-k(i))})$. The point estimate and variance use this same set:

$$\begin{aligned} \hat{\tau}_{\text{agg}} &= \bar{\Phi}^\tau = N_{\text{all}}^{-1} \sum_i \hat{\Phi}_i^\tau, \\ \widehat{\text{Var}}(\hat{\tau}_{\text{agg}}) &= N_{\text{all}}^{-2} \sum_r \sum_{i \in \mathcal{D}_r} (\hat{\Phi}_i^\tau - \bar{\Phi}_r^\tau)^2, \end{aligned}$$

where $\bar{\Phi}_r^\tau$ is the within-site mean. Equivalently, $\hat{V}_\tau = N_{\text{all}} \widehat{\text{Var}}(\hat{\tau}_{\text{agg}})$.

Why within-group centering is necessary. Because $\hat{\Phi}_i^\tau$ is scaled by N_{all}/N_r , its conditional mean can differ across sites. Global centering would therefore add a non-sampling between-site term. The inferential target here is

$$\frac{1}{N_{\text{all}}^2} \sum_{i=1}^{N_{\text{all}}} \text{Var}(\Phi_i^\tau),$$

rather than $\text{Var}(\Phi^\tau)$ under a pooled-mixture distribution. Global centering at $\bar{\Phi}$ estimates the latter object and introduces an additional between-group term

$$\sum_g \rho_g (\mu_g - \mu)^2,$$

equivalently the ANOVA component $\text{BSS}/N_{\text{all}}^2$, which is of the same order as $L_\tau^*(\eta_\tau^*)/N_{\text{all}}$ and therefore does not vanish. For a common fixed weight vector η (or conditionally within one outer fold), $\text{WSS}/N_{\text{all}}^2$ has the familiar decomposition

$$\widehat{\text{Var}}(\hat{\tau}_{\text{agg}}) = \frac{1}{N_t} \hat{\sigma}_{T,\tau}^2 + \sum_{j=1}^K \frac{1}{N_{s_j}} \hat{\eta}_j^2 \hat{\sigma}_{s_j}^2,$$

where

$$\hat{\sigma}_{T,\tau}^2 = \widehat{\text{Var}}_t \left[(1 - \sum_j \hat{\eta}_j) \hat{\varphi}_{ot,i}^\tau + \sum_j \hat{\eta}_j \hat{\zeta}_{t,s_j,i}^\tau \right],$$

and $\hat{\sigma}_{S_j}^2$ is the within-source variance of $\hat{\xi}_{t,s_j,i}^\tau$. Hence $\hat{\sigma}_{T,\tau}^2$ is not a target-only variance term; it already contains the shared-target cross-covariances, including terms corresponding to $\text{Cov}(\hat{\varphi}_{ot}^\tau, \hat{\zeta}_{t,s_j}^\tau)$ and $\text{Cov}(\hat{\zeta}_{t,s_j}^\tau, \hat{\zeta}_{t,s_k}^\tau)$ for $j \neq k$, fully consistent with the $\hat{C}_{ot,s_j}^\tau$ and $\hat{C}_{t,s_j,s_k}^\tau$ terms in Eq. (10).

In the implemented cross-fitted estimator, $\hat{\eta}^{(-k(i))}$ varies across outer evaluation folds, so the final finite-sample WSS is computed directly from the displayed observation-level pseudo-values and is not factored through one global weight vector. Because K_f is fixed and $\max_{k \leq K_f} \|\hat{\eta}^{(-k)} - \eta_\tau^*\|_2 = o_p(1)$, this fold-specific WSS has the same probability limit $L_\tau^*(\eta_\tau^*)$.

With balanced folds ($|\mathcal{D}_{t,k}| = N_t/K_f$, $|\mathcal{D}_{s_j,k}| = N_{s_j}/K_f$), $\bar{\Phi}^\tau$ coincides with the fold-average $(1/K_f) \sum_k \hat{\tau}_{\text{agg},k}$; with minor fold imbalance the difference is $O(N_{\text{all}}^{-1})$.

This unified form ensures $\hat{\tau}_{\text{agg}} - \tau = N_{\text{all}}^{-1} \sum_i (\hat{\Phi}_i^\tau - \tau)$, so the DML representation of the estimation error as an IF average is preserved. By the same arguments as above (nuisance consistency, L_2 -Lipschitz continuity, and $\hat{\eta}^{(-k)} \rightarrow_p \eta_\tau^*$), $\hat{V}_\tau \rightarrow_p L_\tau^*(\eta_\tau^*)$.

S1.4 Pairwise Technical Lemmas

The lemmas in this subsection are ordered by their role in the proof of Theorem S1. Lemma S3 handles the leading influence-function average. Lemma S4 supplies the fold-uniform high-dimensional rates for the calibrated nuisance estimators. Lemma S5 records the population moment equations induced by the calibrated losses, and Lemmas S6 and S7 convert those equations into the second-order bias and plug-in stability bounds used in the main expansion.

Lemma S3 (CLT for IF Average). *Under Assumption S1 and Assumption S2, the influence*

function average is asymptotically normal:

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi_{t,s}(D_i; \vartheta^*) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

Proof. The observations consist of two independent samples: $\{D_i : i \in \mathcal{D}_t\}$ drawn i.i.d. from the target distribution and $\{D_i : i \in \mathcal{D}_s\}$ drawn i.i.d. from the source distribution, with $N_t/N \rightarrow \rho_t \in (0, 1)$ and $N_s/N \rightarrow \rho_s$. Thus the summands $\{\varphi_{t,s}(D_i; \vartheta^*)\}_{i=1}^N$ are independent (but not identically distributed), have mean zero, and have finite $(2 + \delta)$ moments by [Assumption S2\(b\)](#). In particular, the Lindeberg condition holds. We verify a Lyapunov condition, which implies the Lindeberg condition for this independent triangular array. Let $X_{i,N} := \varphi_{t,s}(D_i; \vartheta^*)$ so that $\mathbb{E}[X_{i,N}] = 0$ and $\sum_{i=1}^N \text{Var}(X_{i,N}) = NV_{t,s}^* + o(N)$. By [Assumption S2\(b\)](#), $\sup_{i,N} \mathbb{E}[|X_{i,N}|^{2+\delta}] < \infty$ for some $\delta > 0$. Therefore,

$$\frac{1}{(NV_{t,s}^*)^{1+\delta/2}} \sum_{i=1}^N \mathbb{E}[|X_{i,N}|^{2+\delta}] \leq \frac{CN}{N^{1+\delta/2}} = CN^{-\delta/2} \rightarrow 0,$$

which is the Lyapunov condition. Hence the Lindeberg–Feller CLT yields $\frac{1}{\sqrt{N}} \sum_{i=1}^N X_{i,N} \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*)$, proving the lemma. $\square$

Lemma S4 (Cross-fitted ℓ_1 -penalized nuisance rates (adapted from [Hou et al. \(2025\)](#), Lemma A21)). *Fix a constant number of folds K_f . Let the full sample $\mathcal{D} = \{D_1, \dots, D_N\}$ be partitioned into folds $\mathcal{D} = \bigcup_{k=1}^{K_f} \mathcal{D}_k$, and let $\mathcal{D}^{(-k)} = \mathcal{D} \setminus \mathcal{D}_k$ denote the out-of-fold data. For each fold k , let $\hat{\gamma}^{(-k)}$ be an estimator trained on $\mathcal{D}^{(-k)}$ (hence independent of $\mathcal{D}_k$). For an observation index $i \in \{1, \dots, N\}$, let $\text{fold}(i) \in \{1, \dots, K_f\}$ denote the fold index such that $D_i \in \mathcal{D}_{\text{fold}(i)}$; we use $\hat{\gamma}^{(-\text{fold}(i))}$ to denote the out-of-fold nuisance estimate used when evaluating the loss contribution from D_i . Consider the cross-fitted ℓ_1 -penalized estimator*

$$\hat{\beta} \in \arg \min_{\beta \in \mathbb{R}^p} \left\{ \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}) + \lambda \|\beta\|_1 \right\}, \quad \ell_k(\beta; \gamma) := \tilde{E}_k[\ell(D; \beta, \gamma)],$$

where $\ell(D; \beta, \gamma)$ is convex in β . Define also the fold-averaged empirical risk

$$L_N(\beta) := \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}).$$

Let $\bar{\beta}$ be the population minimizer at a population nuisance target $\bar{\gamma}$,

$$\bar{\beta} \in \arg \min_{\beta \in \mathbb{R}^p} \mathbb{E}[\ell(D; \beta, \bar{\gamma})], \quad s_\beta := \|\bar{\beta}\|_0.$$

Assume the following high-dimensional regularity conditions (cf. Assumption 6 of [Hou et al. \(2025\)](#) and the RSC framework of [Negahban et al. \(2012\)](#)):

- (a) (Sub-Gaussian covariates) Each coordinate of $\phi(\mathbf{X})$ is sub-Gaussian with $\max_{1 \leq j \leq p} \|\phi_j(\mathbf{X})\|_{\psi_2} \leq K_\psi$ (allowing $\|\phi(\mathbf{X})\|_\infty$ to grow at the usual $\sqrt{\log p}$ rate).
- (b) (Response/link tails and smoothness) Y is uniformly sub-Gaussian (or bounded), and the GLM mean function and its derivative are uniformly bounded on a neighborhood of $\bar{\beta}$.
- (c) (Identifiability) The population Gram matrix is non-degenerate:

$$\lambda_{\min}(\mathbb{E}[\phi(\mathbf{X})\phi(\mathbf{X})^T]) \geq c > 0.$$

- (d) (Weight regularity) Any weights induced by γ (e.g., density-ratio type weights) have finite moments and are locally Lipschitz in γ on a neighborhood of $\bar{\gamma}$ (sufficient for the score stability bound used below).
- (e) (Restricted strong convexity) With probability tending to 1, $L_N(\cdot)$ satisfies an RSC inequality on the standard cone associated with $\text{supp}(\bar{\beta})$.
- (f) (Plug-in score domination) For the penalty level used in the generic update, the cross-fitted plug-in score term B defined in the proof satisfies $\Pr\{\|B\|_\infty \leq \lambda/4\} \rightarrow 1$. In the foldwise score-centered special case below, this condition holds with $B = \mathbf{0}$.

In our FACE-HD setting, we impose these regularity conditions in addition to [Assumption S1](#), [Assumption S2](#), and [Assumption S3](#) (noting that (a) matches [Assumption S1\(a\)](#), while (d) is ensured by the overlap/weight-moment assumptions and local smoothness); part (f) records

the plug-in score domination needed by the standard cone argument. Choose $\lambda \asymp \sqrt{\log(p)/N}$.

Then:

Remark S3 (On the choice of N in λ). The scaling $\lambda \asymp \sqrt{\log(p)/N}$ is stated in terms of the total sample size N for notational simplicity. In applications where the nuisance loss for a given arm/site pair is effectively averaged over an arm-specific subsample (e.g., treated units in the source), one may equivalently write $\lambda \asymp \sqrt{\log(p)/N_{\text{eff}}}$ where N_{eff} is the corresponding effective sample size. Under the sampling-fraction regime and overlap/positivity conditions stated above, $N_{\text{eff}} \asymp N$, so the rate statements are unchanged.

1. **(General)**

$$\|\hat{\beta} - \bar{\beta}\|_2 = O_p\left(\sqrt{\frac{s_\beta \log(p)}{N}}\right) + O_p\left(\sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right).$$

2. **(Special)** If, for each fold k , the conditional foldwise score at $\bar{\beta}$ satisfies

$$\mathbb{E}\left[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)}) \mid \mathcal{D}^{(-k)}\right] = \mathbf{0},$$

then the plug-in score term disappears and

$$\|\hat{\beta} - \bar{\beta}\|_2 = O_p\left(\sqrt{\frac{s_\beta \log(p)}{N}}\right).$$

Moreover, we can convert the ℓ_2 -rate to an ℓ_1 -rate by an explicit cone argument. Let $\Delta := \hat{\beta} - \bar{\beta}$ and $\mathcal{A}_\beta := \text{supp}(\bar{\beta})$ with $|\mathcal{A}_\beta| = s_\beta$. From the proof below (see the KKT-based cone inequality leading to [Eq. \(S2\)](#)), on a high-probability event we have

$$\|\Delta_{\mathcal{A}_\beta^c}\|_1 \leq 3\|\Delta_{\mathcal{A}_\beta}\|_1 + \frac{2}{\lambda}\|B\|_\infty\|\Delta\|_1,$$

where $\|B\|_\infty$ is the plug-in score term induced by cross-fitting. By condition (f), $(2/\lambda)\|B\|_\infty \leq 1/2$ w.p.a.1, and hence

$$\|\Delta\|_1 = \|\Delta_{\mathcal{A}_\beta}\|_1 + \|\Delta_{\mathcal{A}_\beta^c}\|_1 \leq 8\|\Delta_{\mathcal{A}_\beta}\|_1 \leq 8\sqrt{s_\beta}\|\Delta\|_2.$$

Combining this with the ℓ_2 bounds above yields the corresponding ℓ_1 -rate:

$$\|\hat{\beta} - \bar{\beta}\|_1 = O_p\left(s_\beta \sqrt{\frac{\log(p)}{N}}\right) + O_p\left(\sqrt{s_\beta} \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right),$$

with the second term dropping in the special case.

Instantiation for FACE-HD calibrated losses. In our application, β is instantiated as either α^1 or γ_1 . If [Assumption S3\(d\)](#) holds (as in the foldwise score-centering condition used by [Hou et al. \(2025\)](#)), then we may invoke the **(Special)** case above and obtain

$$\begin{aligned} \|\hat{\alpha}_{t,s}^{1,(-k)} - \alpha^{1,*}\|_q &= O_p\left(s_\alpha^{1-1/q} \sqrt{\frac{\log(p)}{N}}\right), \\ \|\hat{\gamma}_{s,1}^{(-k)} - \gamma_1^*\|_q &= O_p\left(s_\gamma^{1-1/q} \sqrt{\frac{\log(p)}{N}}\right), \quad q \in \{1, 2\}. \end{aligned}$$

Proof. We give a detailed argument in our notation; it follows the structure of [Hou et al. \(2025\)](#) (proof of Lemma A21). Write the (random) fold-averaged empirical loss as

$$L_N(\beta) := \frac{1}{K_f} \sum_{k=1}^{K_f} \ell_k(\beta; \hat{\gamma}^{(-k)}), \quad \hat{\beta} \in \arg \min_{\beta} \{L_N(\beta) + \lambda \|\beta\|_1\}.$$

Let $\Delta := \hat{\beta} - \bar{\beta}$, and let $\mathcal{A}_\beta := \text{supp}(\bar{\beta})$ with $|\mathcal{A}_\beta| = s_\beta$.

Step 1 (Basic inequality). By optimality,

$$L_N(\bar{\beta} + \Delta) + \lambda \|\bar{\beta} + \Delta\|_1 \leq L_N(\bar{\beta}) + \lambda \|\bar{\beta}\|_1,$$

so

$$L_N(\bar{\beta} + \Delta) - L_N(\bar{\beta}) \leq \lambda(\|\bar{\beta}\|_1 - \|\bar{\beta} + \Delta\|_1) \leq \lambda(\|\Delta_{\mathcal{A}_\beta}\|_1 - \|\Delta_{\mathcal{A}_\beta^c}\|_1).$$

Step 2 (RSC expansion). By convexity,

$$L_N(\bar{\beta} + \Delta) - L_N(\bar{\beta}) \geq \nabla L_N(\bar{\beta})^T \Delta + \mathcal{R}_N(\Delta),$$

where the remainder $\mathcal{R}_N(\Delta)$ satisfies the RSC lower bound ([Assumption S3\(a\)](#) as applied to this loss)

$$\mathcal{R}_N(\Delta) \geq \kappa_1 \|\Delta\|_2^2 - \kappa_2 \frac{\log(p)}{N} \|\Delta\|_1^2$$

on the standard restricted cone. Combining with Step 1 yields

$$\kappa_1 \|\Delta\|_2^2 \leq \|\nabla L_N(\bar{\beta})\|_\infty \|\Delta\|_1 + \lambda(\|\Delta_{\mathcal{A}_\beta}\|_1 - \|\Delta_{\mathcal{A}_\beta^c}\|_1) + \kappa_2 \frac{\log(p)}{N} \|\Delta\|_1^2. \quad (\text{S1})$$

Step 3 (Score decomposition and choice of λ). For each fold, write

$$S_k := \tilde{E}_k[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)})], \quad m_k := \mathbb{E}[S_k \mid \mathcal{D}^{(-k)}].$$

Decompose the fold-averaged score as

$$\nabla L_N(\bar{\beta}) = \underbrace{\frac{1}{K_f} \sum_{k=1}^{K_f} (S_k - m_k)}_{\text{stochastic term } A} + \underbrace{\frac{1}{K_f} \sum_{k=1}^{K_f} m_k}_{\text{plug-in term } B}.$$

For each fixed k , conditional on $\mathcal{D}^{(-k)}$, cross-fitting makes S_k an average of independent sub-exponential random vectors with uniformly controlled coordinates. Bernstein/Hoeffding concentration and a union bound over the coordinates give $\|S_k - m_k\|_\infty = O_p\{\sqrt{\log(p)/N_k}\}$. Because K_f is fixed and $N_k \asymp N$, summing these foldwise bounds (without conditioning simultaneously on the overlapping training samples) yields

$$\|A\|_\infty = O_p\left(\sqrt{\frac{\log(p)}{N}}\right).$$

Choose $\lambda \asymp \sqrt{\log(p)/N}$ so that $\lambda \geq 2\|A\|_\infty$ w.p.a.1.

For the plug-in term, note that

$$B - \mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \bar{\gamma})] = \frac{1}{K_f} \sum_{k=1}^{K_f} \mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)}) - \nabla_{\beta} \ell(D; \bar{\beta}, \bar{\gamma}) \mid \mathcal{D}^{(-k)}],$$

and the weight-regularity/Lipschitz condition implies

$$\|B - \mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \bar{\gamma})]\|_2 \leq C \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2.$$

This L_2 perturbation bound controls the additive plug-in term in the final oracle inequality.

The separate ℓ_∞ domination required for the cone step is imposed in condition (f). In the special case, $\mathbb{E}[\nabla_{\beta} \ell(D; \bar{\beta}, \hat{\gamma}^{(-k)}) \mid \mathcal{D}^{(-k)}] = 0$ for all k , hence $B = \mathbf{0}$.

Step 4 (Cone property and ℓ_2 rate). Plugging $\|\nabla L_N(\bar{\beta})\|_\infty \leq \|A\|_\infty + \|B\|_\infty$ into Eq. (S1) and using $\lambda \geq 2\|A\|_\infty$ yields the cone inequality

$$\|\Delta_{\mathcal{A}_\beta^c}\|_1 \leq 3\|\Delta_{\mathcal{A}_\beta}\|_1 + \frac{2}{\lambda}\|B\|_\infty\|\Delta\|_1. \quad (\text{S2})$$

On the event $\{(2/\lambda)\|B\|_\infty \leq 1/2\}$, Eq. (S2) implies $\|\Delta_{\mathcal{A}_\beta^c}\|_1 \leq 7\|\Delta_{\mathcal{A}_\beta}\|_1$ and hence $\|\Delta\|_1 \leq 8\|\Delta_{\mathcal{A}_\beta}\|_1 \leq 8\sqrt{s_\beta}\|\Delta\|_2$. Plugging this back into the RSC lower bound (Assumption S3(a) as applied to this loss), using $B^T\Delta \leq \|B\|_2\|\Delta\|_2$ for the plug-in score contribution, and absorbing the $\kappa_2(\log(p)/N)\|\Delta\|_1^2$ term yields

$$\|\Delta\|_2 = O_p(\sqrt{s_\beta}\lambda) + O_p\left(\sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right).$$

This is the general-case ℓ_2 rate. In the special case, $B = \mathbf{0}$ and the second term drops.

Step 5 (ℓ_1 rate). From the cone property and $\|\Delta_{\mathcal{A}_\beta}\|_1 \leq \sqrt{s_\beta}\|\Delta\|_2$,

$$\|\Delta\|_1 = O_p(s_\beta\lambda) + O_p\left(\sqrt{s_\beta} \sup_{1 \leq k \leq K_f} \|\hat{\gamma}^{(-k)} - \bar{\gamma}\|_2\right),$$

with the second term dropping in the special case.

Finally, for FACE-HD, the **(Special)** conclusion may be invoked for $\beta = \alpha^1$ and $\beta = \gamma_1$ only when the foldwise score-centering condition is assumed/verified as in Assumption S3(d). Otherwise, one should use the **(General)** bounds, which explicitly propagate the plug-in nuisance error. $\square$

Lemma S5 (Orthogonality from Calibrated Losses). *Under Assumption S3(c), let $\vartheta^* = (\alpha^{1,*}, \gamma_1^*)$ be the population targets of the calibrated losses in the two-level algorithm (see the combined manuscript's fold-level supplement; Eq. (6) and Eq. (7)):*

$$\begin{aligned} L_\gamma(\gamma; \alpha^{1,*}) &= \mathbb{E}_t \left[\nabla_\alpha g(\phi(\mathbf{X}); \alpha^{1,*})^T \gamma \right] + \mathbb{E}_s \left[I(A=1) e^{-\phi(\mathbf{X})^T \gamma} g'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*})) \right] \\ L_\alpha(\alpha; \gamma_1^*) &= \mathbb{E}_s \left[w_\tau(\mathbf{X}; \gamma_1^*) \left(-Y \cdot (\phi(\mathbf{X})^T \alpha) + \Psi(\phi(\mathbf{X})^T \alpha) \right) \right] \end{aligned}$$

(Ignoring regularization terms, which vanish in population). For canonical-link GLMs,

$$\nabla_{\alpha} g(\phi(\mathbf{X}); \alpha) = \phi(\mathbf{X}) g'(\phi(\mathbf{X})^T \alpha).$$

Remark S4 (Penalization and approximate orthogonality). The population moment conditions below are derived from the unpenalized population losses. In finite samples, the ℓ_1 penalty induces a KKT subgradient term of order λ in the estimating equations. Under the nuisance rates in [Lemma S4](#) and the sparsity scaling used in [Theorem S1](#), this penalty-induced perturbation contributes at most an $o_p(N^{-1/2})$ term to the final DR remainder and therefore does not affect the first-order limit theory.

Under asymptotically inactive truncation ([Assumption S2\(c\)](#)), the population first-order optimality conditions are therefore equivalent to the following moment conditions at ϑ^* :

$$\begin{aligned} (i) \quad & \mathbb{E}_t \left[\nabla_{\alpha} g(\phi(\mathbf{X}); \alpha^{1,*}) \right] - \mathbb{E}_s \left[w(\mathbf{X}; \gamma_1^*) \nabla_{\alpha} g(\phi(\mathbf{X}); \alpha^{1,*}) \right] = \mathbf{0} \\ (ii) \quad & \mathbb{E}_s \left[w(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \{Y - g(\phi(\mathbf{X}); \alpha^{1,*})\} \right] = \mathbf{0}. \end{aligned}$$

Proof. We derive the moment conditions by taking the gradients of the population loss functions L_{γ} and L_{α} with respect to the parameter being optimized and setting them to zero at the minimum ϑ^* .

Condition (i) from FOC of $L_{\gamma}(\gamma; \alpha^{1,*})$ w.r.t. γ : The gradient of $L_{\gamma}(\gamma; \alpha^{1,*})$ with respect to γ is:

$$\begin{aligned} \nabla_{\gamma} L_{\gamma}(\gamma; \alpha^{1,*}) &= \mathbb{E}_t \left[\nabla_{\alpha} g(\phi(\mathbf{X}); \alpha^{1,*}) \right] \\ &\quad + \mathbb{E}_s \left[I(A=1) (-\phi(\mathbf{X})) e^{-\phi(\mathbf{X})^T \gamma} g'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*})) \right] \end{aligned}$$

Setting this gradient to zero at $\gamma = \gamma_1^*$ gives:

$$\mathbb{E}_t \left[\nabla_{\alpha} g(\phi(\mathbf{X}); \alpha^{1,*}) \right] = \mathbb{E}_s \left[\phi(\mathbf{X}) w(\mathbf{X}; \gamma_1^*) g'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*})) \right].$$

This equation is constructed to enforce the moment condition (i). Note that for standard GLMs with canonical links,

$$\nabla_{\alpha} g(\phi(\mathbf{X}); \alpha) = \phi(\mathbf{X}) g'(\phi(\mathbf{X})^T \alpha).$$

Truncation Error Analysis: Define the event $\mathcal{E}_{\alpha, N} = \{|\phi(\mathbf{X})^T \alpha^{1,*}| \leq M_{\tau, N}\}$. By [Assumption S2\(c\)](#), $P(\mathcal{E}_{\alpha, N}^c \mid R = s) = o(N^{-1})$. On $\mathcal{E}_{\alpha, N}$, $\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*}) = \phi(\mathbf{X})^T \alpha^{1,*}$, so $g'(\mathcal{T}(\cdot)) = g'(\cdot)$ exactly. Therefore, replacing $g'(\mathcal{T}(\phi(\mathbf{X})^T \alpha^{1,*}))$ by $g'(\phi(\mathbf{X})^T \alpha^{1,*})$ in the population FOC changes the source-side moment by at most an $o(1)$ term, and condition (i) holds asymptotically.

Condition (ii) from FOC of $L_{\alpha}(\alpha; \gamma_1^*)$ w.r.t. α : The loss L_{α} is the population weighted GLM log-likelihood. Its gradient with respect to α is:

$$\nabla_{\alpha} L_{\alpha}(\alpha; \gamma_1^*) = \mathbb{E}_s \left[w_{\tau}(\mathbf{X}; \gamma_1^*) \left(-Y \cdot \phi(\mathbf{X}) + \nabla_{\alpha} \Psi(\phi(\mathbf{X})^T \alpha) \right) \right]$$

Using the GLM identity $\nabla_{\alpha} \Psi(u) = g(u) \phi(\mathbf{X})$ where $u = \phi(\mathbf{X})^T \alpha$:

$$\nabla_{\alpha} L_{\alpha}(\alpha; \gamma_1^*) = \mathbb{E}_s \left[w_{\tau}(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \left(g(\phi(\mathbf{X})^T \alpha) - Y \right) \right]$$

Setting this gradient to zero at $\alpha = \alpha^{1,*}$ gives:

$$\mathbb{E}_s \left[w_{\tau}(\mathbf{X}; \gamma_1^*) \phi(\mathbf{X}) \left(Y - g(\phi(\mathbf{X})^T \alpha^{1,*}) \right) \right] = \mathbf{0}.$$

Finally, under [Assumption S2\(c\)](#), truncation is asymptotically inactive at γ_1^* , so $w_{\tau}(\mathbf{X}; \gamma_1^*) = w(\mathbf{X}; \gamma_1^*)$ w.p.a.1. This yields condition (ii). $\square$

Lemma S6 (Bias-Variance Bounds). *Let $\vartheta_n = (\alpha_n^1, \gamma_{1,n})$ be a sequence of nuisance parameters converging to $\vartheta^* = (\alpha^{1,*}, \gamma_1^*)$.*

1. **(Bias Bound - Double Robustness)** *If ϑ^* minimizes the population calibrated losses ([Lemma S5](#) holds), then:*

$$\begin{aligned} |\mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta_n)] - \mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta^*)]| &\leq C \cdot \left(\|\alpha_n^1 - \alpha^{1,*}\|_2^2 + \|\gamma_{1,n} - \gamma_1^*\|_2^2 \right. \\ &\quad \left. + \|\alpha_n^1 - \alpha^{1,*}\|_2 \cdot \|\gamma_{1,n} - \gamma_1^*\|_2 \right) \end{aligned}$$

2. (Variance Bound)

$$\text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)) \leq C \cdot (\|\alpha_n^1 - \alpha^{1,*}\|_2^2 + \|\gamma_{1,n} - \gamma_1^*\|_2^2)$$

Proof. We analyze the bias term $\mathbb{E}[\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)]$. Let $\Delta_\alpha = \alpha_n^1 - \alpha^{1,*}$ and $\Delta_\gamma = \gamma_{1,n} - \gamma_1^*$. We decompose the bias using an intermediate step:

$$\text{Bias} = \underbrace{\mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha_n^1, \gamma_{1,n}) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_{1,n})]}_{\text{Bias from } \alpha} + \underbrace{\mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_{1,n}) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]}_{\text{Bias from } \gamma}$$

Step 1 (Bias from α : mean-value expansion + orthogonality). Define the path $h_\alpha(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})]$ for $t \in [0, 1]$. By the fundamental theorem of calculus (and dominated convergence, justified by [Assumption S2\(b\)](#) and local boundedness of the derivatives), we have

$$h_\alpha(1) - h_\alpha(0) = \int_0^1 h'_\alpha(t) dt = \int_0^1 \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})]^T \Delta_\alpha dt.$$

Hence

$$\begin{aligned} \text{Bias from } \alpha &= \left(\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \right)^T \Delta_\alpha + \int_0^1 \left\{ \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})] \right. \\ &\quad \left. - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \right\}^T \Delta_\alpha dt. \end{aligned}$$

The first term is zero by [Lemma S5\(i\)](#). For the second term, we use Cauchy–Schwarz:

$$|\text{Bias from } \alpha| \leq \|\Delta_\alpha\|_2 \int_0^1 \left\| \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \right\|_2 dt.$$

The mapping $\vartheta \mapsto \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \vartheta)]$ is locally Lipschitz in ϑ by the explicit derivative formula below, the bounded-slope/Lipschitz condition on g' ([Assumption S2\(b\)](#)), and the bounded-weights/moment conditions ([Assumption S2\(a,b\)](#)). Therefore, for ϑ_n sufficiently close to ϑ^* ,

$$\left\| \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*} + t\Delta_\alpha, \gamma_{1,n})] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)] \right\|_2 \leq C (\|\Delta_\alpha\|_2 + \|\Delta_\gamma\|_2),$$

uniformly over $t \in [0, 1]$, and hence

$$|\text{Bias from } \alpha| \leq C (\|\Delta_\alpha\|_2^2 + \|\Delta_\alpha\|_2 \|\Delta_\gamma\|_2).$$

Step 2 (Bias from γ : mean-value expansion + orthogonality). Similarly define

$h_\gamma(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)]$. Then

$$\text{Bias from } \gamma = \int_0^1 \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)]^T \Delta_\gamma dt.$$

Adding and subtracting $\mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)]$, the leading term is zero by [Lemma S5\(ii\)](#), and the remainder is bounded (again by local Lipschitz continuity and Cauchy–Schwarz) as

$$|\text{Bias from } \gamma| \leq C \|\Delta_\gamma\|_2^2.$$

Step 3 (Collecting). Combining the two displays yields the stated second-order bias bound.

Derivative identities used above. For example, the gradient w.r.t. $\boldsymbol{\alpha}$ at $(\boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)$ is:

$$\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] = \mathbb{E}_t[\nabla_\alpha g(\cdot; \boldsymbol{\alpha}^{1,*})] - \mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \nabla_\alpha g(\cdot; \boldsymbol{\alpha}^{1,*})]$$

where $w(\mathbf{X}; \boldsymbol{\gamma}) = I(A = 1)/e^{\phi(\mathbf{X})^T \boldsymbol{\gamma}}$. This is exactly $\mathbf{0}$ by [Lemma S5\(i\)](#). Similarly, the gradient w.r.t. $\boldsymbol{\gamma}$ is:

$$\begin{aligned} \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] &= \mathbb{E}_s[\nabla_\gamma w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}] \\ &= -\mathbb{E}_s[w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \phi(\mathbf{X}) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}] \end{aligned}$$

This is exactly $\mathbf{0}$ by [Lemma S5\(ii\)](#). This proves part 1.

Part 2 (Variance): explicit L_2 -Lipschitz bound. Write the difference as

$$\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*) = T_t(D) + T_s(D),$$

where

$$T_t(D) := \frac{I(R=t)}{\rho_t} \{g(\phi(\mathbf{X}); \boldsymbol{\alpha}_n^1) - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\},$$

and

$$\begin{aligned} T_s(D) &:= \frac{I(R=s)}{\rho_s} \left(w(\mathbf{X}; \boldsymbol{\gamma}_{1,n}) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}_n^1)\} - w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\} \right) \\ &= \frac{I(R=s)}{\rho_s} \{T_{s,1}(D) + T_{s,2}(D)\}. \end{aligned}$$

where

$$T_{s,1}(D) := \{w(\mathbf{X}; \gamma_{1,n}) - w(\mathbf{X}; \gamma_1^*)\} \{Y - g(\phi(\mathbf{X}); \alpha^{1,*})\},$$

$$T_{s,2}(D) := w(\mathbf{X}; \gamma_{1,n}) \{g(\phi(\mathbf{X}); \alpha^{1,*}) - g(\phi(\mathbf{X}); \alpha_n^1)\}.$$

By $(a + b + c)^2 \leq 3(a^2 + b^2 + c^2)$ and $\text{Var}(Z) \leq \mathbb{E}[Z^2]$,

$$\text{Var}(\tilde{\varphi}_{t,s}(D; \vartheta_n) - \tilde{\varphi}_{t,s}(D; \vartheta^*)) \leq 3 \mathbb{E}[T_t(D)^2] + 3 \mathbb{E}[T_{s,1}(D)^2] + 3 \mathbb{E}[T_{s,2}(D)^2].$$

For T_t and $T_{s,2}$, the bounded-slope assumption on g ([Assumption S2\(b\)](#)) implies $|g(u_1) - g(u_2)| \leq L_g |u_1 - u_2|$, hence

$$\mathbb{E}[T_t(D)^2] + \mathbb{E}[T_{s,2}(D)^2] \leq C \mathbb{E}[\|\phi(\mathbf{X})\|_2^2] \|\Delta_\alpha\|_2^2,$$

using the weight-moment condition ([Assumption S2\(a\)](#)) and finite second moments of $\phi(\mathbf{X})$ ([Assumption S1\(a\)](#)). For $T_{s,1}$, local L_2 -Lipschitz continuity of $w(\mathbf{X}; \gamma)$ in γ (implied by the exponential form of w , the overlap/weight-moment assumptions, and the moment condition in [Assumption S2\(b\)](#)) yields

$$\mathbb{E}[T_{s,1}(D)^2] \leq C \|\Delta_\gamma\|_2^2.$$

Combining the bounds gives the stated variance inequality. $\square$

Lemma S7 (Double Robustness of Bias Terms). *Let $\Delta_\alpha = \alpha_n^1 - \alpha^{1,*}$ and $\Delta_\gamma = \gamma_{1,n} - \gamma_1^*$.*

Define the two first-order bias components:

$$\mathcal{B}_\alpha(\alpha_n^1; \gamma_1^*) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha_n^1, \gamma_1^*) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]$$

$$\mathcal{B}_\gamma(\gamma_{1,n}; \alpha^{1,*}) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_{1,n}) - \tilde{\varphi}_{t,s}(D; \alpha^{1,*}, \gamma_1^*)]$$

Under the conditions:

1. *If [Lemma S5\(i\)](#) holds (from calibration w.r.t. γ), then $\mathcal{B}_\alpha(\alpha_n^1; \gamma_1^*) = O(\|\Delta_\alpha\|_2^2)$.*
2. *If [Lemma S5\(ii\)](#) holds (from calibration w.r.t. α), then $\mathcal{B}_\gamma(\gamma_{1,n}; \alpha^{1,*}) = O(\|\Delta_\gamma\|_2^2)$.*

Note: For identification of the target parameter, [Theorem S1](#) still requires at least one nuisance model to be correctly specified so that $\mu^1(\vartheta^) = \mu^1$. This is formalized in [Lemma S1](#).*

Proof. 1. Analyzing $\mathcal{B}_\alpha$ using Lemma S5(i) (mean-value expansion). Let $h_1(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)]$ for $t \in [0, 1]$. Then $\mathcal{B}_\alpha = h_1(1) - h_1(0)$ and

$$\mathcal{B}_\alpha = \int_0^1 h_1'(t) dt = \int_0^1 \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)]^T \Delta_\alpha dt.$$

Add and subtract $\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)]$ inside the integral to get

$$\begin{aligned} \mathcal{B}_\alpha &= \left(\mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right)^T \Delta_\alpha + \int_0^1 \left\{ \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)] \right. \\ &\quad \left. - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\}^T \Delta_\alpha dt. \end{aligned}$$

The first term is zero by Lemma S5(i). For the second term, we bound the gradient difference. From the definition of $\tilde{\varphi}_{t,s}$ and the canonical-link GLM identity $\nabla_\alpha g(\phi(\mathbf{X}); \boldsymbol{\alpha}) = \phi(\mathbf{X})g'(\phi(\mathbf{X})^T \boldsymbol{\alpha})$,

$$\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}, \boldsymbol{\gamma}_1^*) = \frac{I(R=t)}{\rho_t} \phi(\mathbf{X})g'(\phi(\mathbf{X})^T \boldsymbol{\alpha}) - \frac{I(R=s)}{\rho_s} w(\mathbf{X}; \boldsymbol{\gamma}_1^*) \phi(\mathbf{X})g'(\phi(\mathbf{X})^T \boldsymbol{\alpha}).$$

Using the Lipschitz property of g' (Assumption S2(b)) and the weight-moment condition (Assumption S2(a)), we have

$$\left\| \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*} + t\Delta_\alpha, \boldsymbol{\gamma}_1^*)] - \mathbb{E}[\nabla_\alpha \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\|_2 \leq C \|\Delta_\alpha\|_2,$$

uniformly over $t \in [0, 1]$, where $C < \infty$ depends on ρ_t, ρ_s , the moment bound on $\phi(\mathbf{X})$ (Assumption S1(a)), and $L_{g'}$. Therefore, by Cauchy-Schwarz,

$$|\mathcal{B}_\alpha| \leq \int_0^1 C \|\Delta_\alpha\|_2 \cdot \|\Delta_\alpha\|_2 dt = O(\|\Delta_\alpha\|_2^2).$$

2. Analyzing $\mathcal{B}_\gamma$ using Lemma S5(ii) (mean-value expansion). Let $h_2(t) := \mathbb{E}[\tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)]$. Then $\mathcal{B}_\gamma = h_2(1) - h_2(0)$ and

$$\mathcal{B}_\gamma = \int_0^1 \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)]^T \Delta_\gamma dt.$$

Add and subtract the gradient at $\boldsymbol{\gamma}_1^*$:

$$\begin{aligned} \mathcal{B}_\gamma &= \left(\mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right)^T \Delta_\gamma + \int_0^1 \left\{ \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^* + t\Delta_\gamma)] \right. \\ &\quad \left. - \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \boldsymbol{\gamma}_1^*)] \right\}^T \Delta_\gamma dt. \end{aligned}$$

The first term is zero by [Lemma S5\(ii\)](#). For the remainder, recall $w(\mathbf{X}; \gamma) = I(A = 1) \exp(-\phi(\mathbf{X})^T \gamma)$ so that $\nabla_\gamma w(\mathbf{X}; \gamma) = -w(\mathbf{X}; \gamma) \phi(\mathbf{X})$. Thus

$$\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma) = \frac{I(R=s)}{\rho_s} \nabla_\gamma w(\mathbf{X}; \gamma) \{Y - g(\phi(\mathbf{X}); \boldsymbol{\alpha}^{1,*})\}.$$

Under [Assumption S2\(a,b\)](#), the mapping $\gamma \mapsto \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma)]$ is locally Lipschitz in γ , hence

$$\left\| \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma_1^* + t\Delta_\gamma)] - \mathbb{E}[\nabla_\gamma \tilde{\varphi}_{t,s}(D; \boldsymbol{\alpha}^{1,*}, \gamma_1^*)] \right\|_2 \leq C \|\Delta_\gamma\|_2$$

uniformly over $t \in [0, 1]$. Therefore $|\mathcal{B}_\gamma| \leq \int_0^1 C \|\Delta_\gamma\|_2^2 dt = O(\|\Delta_\gamma\|_2^2)$.

Clarification on Double Robustness: The key insight is that the calibrated loss functions are designed such that:

- [Lemma S5\(i\)](#) holds *by construction* of L_γ at any ϑ^* that minimizes L_γ , regardless of whether the weighting model is correct.
- [Lemma S5\(ii\)](#) holds *by construction* of L_α at any ϑ^* that minimizes L_α , regardless of whether the outcome model is correct.

Thus the first-order sensitivity to each nuisance can be removed by calibration; combined with model correctness of at least one nuisance for identification, this yields the double-robust conclusion in [Theorem S1](#). $\square$

S1.5 Extension to the Control Mean and the TATE

The results above were stated for the treated potential outcome mean $\mu^1 = \mathbb{E}_t[Y(1)]$ for notational simplicity. The same proof strategy applies to the control mean μ^0 by replacing the treatment level 1 by 0 throughout (including the corresponding nuisance parameters $(\boldsymbol{\alpha}^0, \gamma_0)$ and weights). In particular, define the control-arm weight

$$w^0(\mathbf{X}; \gamma_0) := \frac{I(A=0)}{e^{\phi(\mathbf{X})^T \gamma_0}},$$

and the cross-fitted estimator

$$\hat{\mu}_{t,s}^{0,\text{CF}} = \frac{1}{K_f} \sum_{k=1}^{K_f} \left\{ \tilde{E}_{t,k} \left[g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t,s}^{0,(-k)}) \right] + \tilde{E}_{s,k} \left[w^0(\mathbf{X}; \hat{\boldsymbol{\gamma}}_{s,0}^{(-k)}) (Y - g(\phi(\mathbf{X}); \hat{\boldsymbol{\alpha}}_{t,s}^{0,(-k)})) \right] \right\}.$$

Let $\vartheta^{0,*} = (\boldsymbol{\alpha}^{0,*}, \boldsymbol{\gamma}_0^*)$ denote the population targets of the control-arm calibrated losses (the same construction as for μ^1 but with $I(A=0)$), and define the corresponding centered influence function $\varphi_{t,s}^0(D; \vartheta^{0,*})$ analogously.

Theorem S2 (Asymptotic Normality of $\hat{\mu}_{t,s}^{0,\text{CF}}$). *Under the analogues of [Assumptions S1](#) to [S3](#) for the control arm, and assuming identification of $\mu^0(\vartheta^{0,*}) = \mu^0$ under outcome-model or weight-model correctness (control-arm analogue of [Lemma S1](#)), if the total sparsity satisfies $\frac{s_{\text{tot}}^2 \log^2(p)}{N} \rightarrow 0$, then*

$$\sqrt{N} \left(\hat{\mu}_{t,s}^{0,\text{CF}} - \mu^0 \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^{0,*}), \quad V_{t,s}^{0,*} := \text{Var} \left(\varphi_{t,s}^0(D; \vartheta^{0,*}) \right).$$

Proof. The proof is identical to that of [Theorem S1](#) after replacing the treatment level 1 by 0 in the definition of the score contribution, the calibrated moment conditions, and the corresponding nuisance estimators. $\square$

Corollary S1 (Asymptotic Normality of the TATE). *With $\tau = \mu^1 - \mu^0$ as defined above, define the cross-fitted TATE estimator*

$$\hat{\tau}_{t,s}^{\text{CF}} := \hat{\mu}_{t,s}^{1,\text{CF}} - \hat{\mu}_{t,s}^{0,\text{CF}}.$$

Under the conditions of [Theorem S1](#) and [Theorem S2](#), we have the asymptotic linear representation

$$\sqrt{N} \left(\hat{\tau}_{t,s}^{\text{CF}} - \tau \right) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \left(\varphi_{t,s}(D_i; \vartheta^*) - \varphi_{t,s}^0(D_i; \vartheta^{0,*}) \right) + o_p(1),$$

and hence

$$\sqrt{N} \left(\hat{\tau}_{t,s}^{\text{CF}} - \tau \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^{\tau,*}), \quad V_{t,s}^{\tau,*} := \text{Var} \left(\varphi_{t,s}(D; \vartheta^*) - \varphi_{t,s}^0(D; \vartheta^{0,*}) \right).$$

The corollary gives a sufficient route through separate arm identification. For aggregation, a weaker estimand-level route is possible. Let $\mu_{t,s}^{a,*} = \text{plim } \hat{\mu}_{t,s}^{a,\text{CF}}$. If $\mu_{t,s}^{1,*} - \mu_{t,s}^{0,*} = \tau$ and the centered arm contrast admits the same influence-function expansion with an $o_p(N^{-1/2})$ remainder, then the source is first-order valid for the TATE even when neither arm-specific pseudo-limit equals μ^a . The contrast expansion is imposed explicitly in [Assumption S4\(e\)](#); equality of the two pseudo-limit biases alone is not sufficient.

S2 Additional Simulation Results

The direct-TATE rerun uses $p = 100$ for C(1)–C(3). The archived figures below report the preceding treated-mean results at $p = 50$: C(2) omits the quadratic outcome terms, while C(3) omits the quadratic site/propensity terms. Because the skew-normal source-to-target density ratio is not exactly log-quadratic, C(1) is a rich working-model configuration rather than literal joint parametric correctness. These figures are retained for reproducibility and are not part of the new direct-TATE simulation array. They show that FACE-HD is markedly more stable than the non-adaptive aggregators as source heterogeneity grows, but they should not be interpreted as showing the lowest RMSE at every value of ρ or nominal coverage in every C(3) setting. Direct-TATE rerun figures at $p = 100$ are reported separately using the same seeds and data-generating settings.

[Table S1](#) records the bounded pilot used to lock the production Wald penalty-activation cutoff. The pilot used ten seeds (501–510) disjoint from production seeds 1–500 and did not use coverage in the selection rule. All candidates satisfied the no-deviation efficiency screen, and $c = 1$ minimized the worst RMSE at $\rho \in \{1, 2.5\}$. These ten-run values are a tuning audit rather than final performance estimates; the 500-replicate cutoff sensitivity is reported with the production analysis.

To isolate the C(3) centering mechanism before adaptive source aggregation, we also ran

Table S1: Coverage-blind cutoff-locking pilot for $C(1)$, $p = 100$, and $K = 2$. Each RMSE uses ten disjoint pilot replicates. “Eligible” means that the $\rho = 0$ RMSE is at most 1.05 times the smallest value in that column.

Cutoff c	Multiplier λ	RMSE ($\rho = 0$)	RMSE ($\rho = 1$)	RMSE ($\rho = 2.5$)	Eligible/selected
1.0	1.000	0.01957	0.03105	0.02938	Yes/Yes
1.5	0.667	0.01936	0.03623	0.03847	Yes/No
2.0	0.500	0.01941	0.04239	0.05061	Yes/No
2.5	0.400	0.01959	0.04677	0.06400	Yes/No
3.0	0.333	0.01970	0.04964	0.07414	Yes/No

the target-only TATE estimator for 500 replicates at $p = 100$, $K = 4$, and $\rho = 0$, replacing either nuisance prediction by its known DGP value on each held-out fold. [Table S2](#) reports the result. The fitted–fitted estimator has a material negative bias even though its standard error is slightly conservative relative to the empirical spread. Either oracle nuisance removes nearly all of that bias and restores nominal coverage, which is the finite-sample signature of a doubly robust product remainder. This focused experiment diagnoses the target anchor; it is not substituted for the full direct-TATE comparison array.

Because the four variants use the same replicate and folds, their paired inclusion–exclusion contrast isolates the interaction of the two nuisance errors. Specifically, fitted/fitted minus oracle-outcome minus oracle-propensity plus oracle/oracle has mean -0.0159 and Monte Carlo SE 0.0003 . This accounts for nearly all of the fitted/fitted centering bias and supports the product-remainder diagnosis rather than a variance-scale failure. Under the one-standard-error target rule, the same interaction remainder is -0.0231 (Monte Carlo SE 0.0003). Its paired difference from the primary rule is -0.00721 (Monte Carlo SE 0.00017), while the paired fitted/fitted estimate shift is -0.00723 (Monte Carlo SE 0.00024). The double-oracle rows agree exactly replicate by replicate, so the deterioration is not a change in

Table S2: C(3) target-only TATE oracle-remainder diagnostic, based on 500 paired replicates at $p = 100$, $K = 4$, and $\rho = 0$. SD is the empirical Monte Carlo standard deviation and mean SE is the average influence-function standard error. The primary simulation analysis uses the minimum-validation-loss rule; the lower panel changes only the fitted target nuisance rule.

Nuisance predictions	Bias	SD	Mean SE	RMSE	Coverage
<i>Minimum-validation-loss rule (primary)</i>					
Fitted outcome, fitted propensity	−0.0178	0.0290	0.0302	0.0341	0.918
Oracle outcome, fitted propensity	−0.0021	0.0279	0.0290	0.0280	0.952
Fitted outcome, oracle propensity	−0.0018	0.0319	0.0324	0.0319	0.952
Oracle outcome, oracle propensity	−0.0019	0.0307	0.0311	0.0307	0.950
<i>One-standard-error rule (target-nuisance sensitivity)</i>					
Fitted outcome, fitted propensity	−0.0251	0.0292	0.0302	0.0384	0.872
Oracle outcome, fitted propensity	−0.0022	0.0275	0.0287	0.0275	0.954
Fitted outcome, oracle propensity	−0.0018	0.0324	0.0328	0.0324	0.958
Oracle outcome, oracle propensity	−0.0019	0.0307	0.0311	0.0307	0.950

the generated data, folds, or truth. Stronger target regularization therefore increases rather than removes the finite-sample product remainder; the primary simulation minimum-loss rule is retained without coverage-based retuning.

The $\rho = 0$ endpoint fixes the conditional treatment log-odds shift at one in every site, but it does not force the raw site-specific marginal risk differences to be identical. [Table S3](#) evaluates those true marginal effects directly under each site’s covariate law. This separates the small covariate-shift/non-collapsibility discrepancy from nuisance estimation and adaptive weighting, and helps explain why SS or IVW can exchange a small centering error for a larger variance reduction at this endpoint. The four source-law treated means exceed the target

Table S3: True site-marginal TATEs at $\rho = 0$, $p = 100$, and $K = 4$.

Site	Skewness ν_r	Marginal TATE	Difference from target
Target	0.00	0.2063	0.0000
s_1	0.05	0.2037	−0.0026
s_2	0.10	0.2014	−0.0048
s_3	0.15	0.1993	−0.0070
s_4	0.20	0.1970	−0.0093

$\mu^1 = 0.6927$ by 0.0079, 0.0153, 0.0218, and 0.0292, while their control-mean differences are 0.0105, 0.0202, 0.0288, and 0.0384; subtracting the arms partially cancels these shifts and yields the smaller TATE differences in the table. The calculation uses 500,000 reference draws per site, with every integration Monte Carlo SE for a TATE contrast below 9×10^{-5} .

Legacy lower-dimensional setting ($p = 50$). The archived treated-mean study used the lower dimension $p = 50$ (with $n = 1000$ observations per site, so $N = 1000(K + 1)$ in total). [Figure S3](#) shows C(1), which uses quadratic working bases for both nuisances. At $\rho = 0$, all sites share the conditional outcome mechanism and treatment log-odds shift. A non-adaptive estimator may attain smaller finite-sample RMSE when its variance reduction outweighs the modest covariate-shift-induced differences in site-specific marginal effects, and no adaptation is needed. As ρ increases, FACE-HD gives a substantially more stable bias–coverage tradeoff, whereas SS, IVW, Federated-DR, and Pooled-DR become biased and lose coverage. The C(2)–C(3) figures at $p = 50$ display the same qualitative robustness pattern, with some finite-sample undercoverage under C(3).

One-round versus two-round. [Figure S4](#) compares the archived arm-wise TATE implementations of one-round and two-round FACE-HD in a continuous-outcome setting where the sources remain mean-compatible but their conditional outcome models differ in shape

from the target’s—a source-specific effect modification of strength s , centered so that each source’s target-population mean treatment effect is unchanged. Across the full range of s and at both $p = 10$ and $p = 50$, the one-round and two-round curves track one another closely in bias, RMSE, coverage, and interval length, and both are more efficient than the target-only anchor. This is descriptive evidence rather than a formal equivalence test: the archived cells contain 80 completed replicates at $p = 10$ and 52–60 at $p = 50$. It supports adopting the communication-light one-round version as the default initialization; the two-round variant remains available as an alternative that relaxes the target-initialization alignment condition. This diagnostic predates the common-weight direct-TATE aggregation and is retained without relabeling it as the new estimator.

S3 Aggregated Estimator: FACE-HD

This section develops the aggregation theory for the direct-TATE estimator in the main manuscript. Arm-specific nuisances are fitted first, but their influence-function components are contrasted before weights, Wald statistics, or variances are calculated. The main result is [Theorem S3](#).

The proof proceeds in five steps. First, [Lemma S8](#) gives the quadratic TATE variance $L_\tau^*(\eta)$. Second, [Lemma S9](#) establishes uniform convergence of its empirical analogue. Third, [Lemma S11](#) proves TATE-level source selection. Fourth, [Lemma S12](#) makes estimated and oracle weights first-order equivalent. Finally, the multisite CLT and Slutsky’s theorem give valid TATE inference.

S3.1 Setup and Notation for Aggregation

Following [Eq. \(9\)](#), aggregate the source-assisted TATE estimators $\{\hat{\tau}_{t,s_j}\}_{j=1}^K$ with the target-only TATE $\hat{\tau}_{ot}$ using one common weight vector. Define the global sample size across all

participating sites as

$$N_{\text{all}} := N_t + \sum_{j=1}^K N_{s_j},$$

and assume $N_t/N_{\text{all}} \rightarrow \rho_t^{\text{all}} \in (0, 1)$ and $N_{s_j}/N_{\text{all}} \rightarrow \rho_{s_j}^{\text{all}} \in (0, 1)$ for each fixed j .

Notation correspondence with the main manuscript:

- $\hat{\tau}_{ot} = \hat{\mu}_{ot}^1 - \hat{\mu}_{ot}^0$: target-only TATE;
- $\hat{\tau}_{t,s_j} = \hat{\mu}_{t,s_j}^1 - \hat{\mu}_{t,s_j}^0$: source-assisted TATE from s_j ;
- $\varphi_{ot,i}^\tau = \varphi_{ot,i}^1 - \varphi_{ot,i}^0$, $\zeta_{j,i}^\tau = \zeta_{j,i}^1 - \zeta_{j,i}^0$, and $\xi_{j,i}^\tau = \xi_{j,i}^1 - \xi_{j,i}^0$: treatment-contrast influence components.

The direct-TATE estimator is

$$\hat{\tau}_{\text{agg}}(\eta) = \hat{\tau}_{ot} + \sum_{j=1}^K \eta_j (\hat{\tau}_{t,s_j} - \hat{\tau}_{ot}).$$

Lemma S8 (Variance-Covariance Structure for Aggregation). *For each source site s_j , let $\varphi_{t,s_j,i}^\tau$ denote the population influence contribution of $\hat{\tau}_{t,s_j}$ under common $\sqrt{N_{\text{all}}}$ scaling, and let $\varphi_{ot,i}^\tau$ denote the target-only TATE contribution. Write*

$$\varphi_{t,s_j,i}^\tau = \zeta_{j,i}^\tau + \xi_{j,i}^\tau.$$

Then, for $j \neq k$,

$$\text{Cov}(\xi_j^\tau, \xi_k^\tau) = 0, \quad \text{Cov}(\zeta_j^\tau, \xi_k^\tau) = 0, \quad \text{Cov}(\varphi_{ot}^\tau, \xi_k^\tau) = 0,$$

because these terms are based on disjoint source samples or on target-versus-source samples that are independent across sites. Consequently, all cross-site covariance terms arise only through the shared target component:

$$\text{Cov}(\varphi_{t,s_j}^\tau, \varphi_{t,s_k}^\tau) = \text{Cov}(\zeta_j^\tau, \zeta_k^\tau), \quad \text{Cov}(\varphi_{ot}^\tau, \varphi_{t,s_j}^\tau) = \text{Cov}(\varphi_{ot}^\tau, \zeta_j^\tau).$$

In particular, the population variance of the aggregated estimator can be written as a quadratic form

$$L_\tau^*(\eta) = \eta^T H_\tau \eta + \mathbf{d}_\tau^T \eta + c_\tau,$$

where the off-diagonal entries of H are induced entirely by the shared target sample.

Proof. The target sample is independent of every source sample, and source samples are mutually independent. Taking an arm contrast changes neither this independence nor the site on which a component is evaluated. Hence source–source and target–source covariances vanish as displayed. Expanding the variance of

$$\varphi_{\text{agg}}^\tau(\eta) = \varphi_{ot}^\tau + \sum_{j=1}^K \eta_j (\varphi_{t,s_j}^\tau - \varphi_{ot}^\tau)$$

then gives the stated quadratic representation, with off-diagonal terms coming solely from the shared target contribution. $\square$

Remark S5 (Relation to FACE as a special case). [Lemma S8](#) is compatible with the covariance decomposition underlying [Han et al. \(2025\)](#). In particular, if the source-assisted estimators are specialized to the FACE construction and the nuisance structure is simplified to the corresponding low-dimensional setting, then the same argument shows that cross-source covariance is induced only by the shared target sample. Thus the FACE variance formula can be viewed as a special case of the present decomposition, whereas the current statement is written to accommodate the more general high-dimensional calibrated nuisance estimators used in FACE-HD.

S3.2 Penalized Variance Estimation

Define the empirical direct-TATE loss:

$$\widehat{L}_\tau(\eta) = N_{\text{all}} \widehat{\text{Var}}\{\widehat{\tau}_{\text{agg}}(\eta)\}$$

which is a quadratic function of η (equivalently, N_{all} times the variance expression in [Eq. \(10\)](#) in the main manuscript):

$$\widehat{L}_\tau(\eta) = \eta^T \widehat{H}_\tau \eta + \widehat{\mathbf{d}}_\tau^T \eta + \widehat{c}_\tau$$

where every component is calculated after taking the arm contrast.

For the implemented two-level cross-fitting, let $\widehat{L}_{\tau,-k}$ denote this objective calculated only from the outer training sample $\mathcal{T}_k$, and let $\widehat{\eta}^{(-k)}$ be its penalized minimizer. In the supporting arguments below, $\widehat{L}_\tau$ and $\widehat{\eta}$ are shorthand for an arbitrary fixed outer fold's $\widehat{L}_{\tau,-k}$ and $\widehat{\eta}^{(-k)}$. Since K_f is fixed, every $o_p(1)$ statement holds uniformly over $k \leq K_f$. The final held-out pseudo-value variance $\widehat{V}_\tau$ is a distinct finite-sample quantity; both it and the foldwise training objectives evaluated at their minimizers converge to $L_\tau^*(\eta_\tau^*)$.

Lemma S9 (Consistency of Variance/Covariance Components). *Assume [Assumptions S1](#) to [S3](#) for both arms, and let K be fixed. Let $H_\tau, \mathbf{d}_\tau, c_\tau$ denote the population analogues. Define*

$$a_N := \max_{a \in \{0,1\}} \max_{j \leq K} \max_{k \leq K_f} \left\{ \|\widehat{\boldsymbol{\alpha}}_{t,s_j}^{a,(-k)} - \boldsymbol{\alpha}_{t,s_j}^{a,*}\|_2 + \|\widehat{\boldsymbol{\gamma}}_{s_j,a}^{(-k)} - \boldsymbol{\gamma}_{s_j,a}^*\|_2 \right\} = o_p(1).$$

Then:

$$\|\widehat{H}_\tau - H_\tau\|_{\text{op}} + \|\widehat{\mathbf{d}}_\tau - \mathbf{d}_\tau\|_2 + |\widehat{c}_\tau - c_\tau| = o_p(1),$$

and consequently, for any fixed compact set $\mathcal{B} \subset \mathbb{R}^K$,

$$\sup_{\eta \in \mathcal{B}} |\widehat{L}_\tau(\eta) - L_\tau^*(\eta)| = o_p(1).$$

If the products of the relevant influence-function components have uniformly bounded second moments, the empirical-averaging part is $O_p(N_{\text{all}}^{-1/2})$, giving the sharper bound $O_p(N_{\text{all}}^{-1/2}) + O_p(a_N)$. This sharper rate is not needed for the FACE-HD oracle limit.

Proof. For readability in this proof only, suppress the subscript τ on $\widehat{H}_\tau, H_\tau, \widehat{\mathbf{d}}_\tau, \mathbf{d}_\tau, \widehat{c}_\tau, c_\tau, \widehat{L}_\tau$, and L_τ^* . We prove entrywise consistency and then convert to the stated matrix/vector norms using that K is fixed.

Step 1 (Generic moment form). Each element of $\widehat{H}$, $\widehat{\mathbf{d}}$ and $\widehat{c}$ is a (finite) linear combination of empirical averages of the form

$$\widehat{M}_r = \frac{1}{N_r} \sum_{i \in \mathcal{D}_r} f(D_i; \widehat{\vartheta}^{(-\text{fold}(i))}), \quad r \in \{t, s_1, \dots, s_K\},$$

where f is a product of two empirical influence-function components (including the cross-site covariance terms computed on the *same* target sample), and $\widehat{\vartheta}^{(-\text{fold}(i))}$ denotes the cross-fitted nuisance estimate used to evaluate the fold containing i . Let M_r be the population analogue obtained by replacing the sample average with $\mathbb{E}_r[\cdot]$ and $\widehat{\vartheta}^{(-\text{fold}(i))}$ with the population target ϑ^* .

Step 2 (Conditional LLN and plug-in stability). Partition the site- r average by evaluation fold:

$$\widehat{M}_r = \sum_{k=1}^{K_f} \frac{N_{r,k}}{N_r} \widehat{M}_{r,k}, \quad \widehat{M}_{r,k} = \frac{1}{N_{r,k}} \sum_{i \in \mathcal{D}_{r,k}} f(D_i; \widehat{\vartheta}^{(-k)}).$$

For each fixed k , conditional on $\mathcal{D}^{(-k)}$, the nuisance estimate is fixed and the evaluation-fold summands in $\widehat{M}_{r,k}$ are independent. The conditional weak law of large numbers therefore gives

$$\widehat{M}_{r,k} - \mathbb{E}_r[f(D; \widehat{\vartheta}^{(-k)}) \mid \mathcal{D}^{(-k)}] = o_p(1).$$

Because K_f is fixed, summing these foldwise statements gives

$$\widehat{M}_r - \sum_{k=1}^{K_f} \frac{N_{r,k}}{N_r} \mathbb{E}_r[f(D; \widehat{\vartheta}^{(-k)}) \mid \mathcal{D}^{(-k)}] = o_p(1).$$

This conditions on one fold's training sample at a time, rather than jointly on the overlapping collection $\{\mathcal{D}^{(-k)}\}_{k=1}^{K_f}$. If the products have uniformly bounded second moments, each foldwise empirical term is $O_p(N_{r,k}^{-1/2})$, and the fixed weighted sum is $O_p(N_r^{-1/2})$. [Assumption S2\(b\)](#) gives the required uniform integrability in a neighborhood of ϑ^* . Next, by [Assumption S3\(d\)](#), $\widehat{\vartheta}^{(-k)} \rightarrow_p \vartheta^*$, and by the same local L_2 stability argument used in [Lemma S6](#) (Part 2), the mapping $\vartheta \mapsto f(D; \vartheta)$ is locally L_2 -Lipschitz. Therefore,

$$\max_{1 \leq k \leq K_f} \left| \mathbb{E}_r[f(D; \widehat{\vartheta}^{(-k)}) \mid \mathcal{D}^{(-k)}] - \mathbb{E}_r[f(D; \vartheta^*)] \right| = O_p(a_N),$$

and combining the two displays yields $\widehat{M}_r - M_r = o_p(1) + O_p(a_N) = o_p(1)$. Under the additional second-moment condition above, the empirical-averaging part is $O_p(N_r^{-1/2})$, hence $O_p(N_{\text{all}}^{-1/2})$ under the site-ratio condition $N_r/N_{\text{all}} \rightarrow \rho_r > 0$.

Step 3 (From entrywise to matrix/vector norms). Since $\widehat{H}$ has fixed dimension $K \times K$ and $\widehat{\mathbf{d}}$ has fixed dimension K , the preceding bound applies to each entry, hence

$$\|\widehat{H} - H\|_{\text{op}} \leq C_K \max_{j,k} |\widehat{H}_{jk} - H_{jk}| = o_p(1), \quad \|\widehat{\mathbf{d}} - \mathbf{d}\|_2 \leq \sqrt{K} \max_j |\widehat{d}_j - d_j| = o_p(1),$$

Finally, $\widehat{c}$ is a finite linear combination of the same sample averages $\{\widehat{M}_r\}$ (coming from the constant terms in the variance expansion), so the same entrywise bound implies $|\widehat{c} - c| = o_p(1)$.

Step 4 (Uniform convergence of the quadratic form). For η in a fixed compact set $\mathcal{B} \subset \mathbb{R}^K$,

$$|\widehat{L}(\eta) - L^*(\eta)| \leq \|\widehat{H} - H\|_{\text{op}} \|\eta\|_2^2 + \|\widehat{\mathbf{d}} - \mathbf{d}\|_2 \|\eta\|_2 + |\widehat{c} - c|,$$

and the right-hand side is $o_p(1)$ uniformly over $\eta \in \mathcal{B}$ because $\sup_{\eta \in \mathcal{B}} \|\eta\|_2 < \infty$. $\square$

The penalized objective uses the TATE-level truncated-Wald penalty. Recall $\widehat{L}_\tau(\eta) = N_{\text{all}} \widehat{\text{Var}}\{\widehat{\tau}_{\text{agg}}(\eta)\}$:

$$\widehat{\eta} = \arg \min_{\eta \in \mathbb{R}^K} \left\{ \widehat{L}_\tau(\eta) + \sum_{j=1}^K (\lambda \widehat{t}_j^\tau - 1)_+ |\eta_j| \right\}, \quad \widehat{t}_j^\tau := \frac{|\widehat{\tau}_{ot} - \widehat{\tau}_{t,s_j}|}{\widehat{\text{SE}}(\text{disc}_j^\tau)}, \quad (\text{S3})$$

where $\text{disc}_j^\tau = \widehat{\tau}_{ot} - \widehat{\tau}_{t,s_j}$ and

$$\widehat{\text{SE}}(\text{disc}_j^\tau)^2 = \frac{\widehat{V}_{ot}^\tau}{N_t} + \frac{\widehat{V}_{t,s_j}^\tau}{N_t} + \frac{\widehat{V}_{s_j}^\tau}{N_{s_j}} - \frac{2\widehat{C}_{ot,s_j}^\tau}{N_t} = \widehat{\text{Var}}(\widehat{\tau}_{t,s_j} - \widehat{\tau}_{ot}).$$

Equivalently, the implementation minimizes the unscaled variance plus $N_{\text{all}}^{-1} \sum_j (\lambda \widehat{t}_j^\tau - 1)_+ |\eta_j|$.

This exact rescaling is tested in the codebase. The penalty is inactive when $\widehat{t}_j^\tau \leq 1/\lambda$. The oracle-selection regime is $1/\lambda \rightarrow \infty$ while $1/\lambda = o(\sqrt{N_{\text{all}}} b_{0,N}^\tau)$.

Remark S6 (FACE-style validation for λ). The main manuscript fixes the aggregation penalty-activation threshold at the locked pilot value $1/\lambda = 1$ and does not tune it on the analyzed data. The FACE sample-splitting validation rule described here is an optional alternative for those who instead wish to tune λ over a grid: for each outer evaluation fold, candidate weights are learned on inner-training summaries by solving the penalized aggregation problem, and these fixed weights are then evaluated on held-out inner-validation summaries through the pure validation-variance criterion. The penalty is used to construct each training-fold candidate but is not added again to its held-out validation score. This optional validation step is a finite-sample rule for choosing a stable point along the penalized aggregation path; it is not the same object as the nuisance-model cross-validation used for α and γ . For the asymptotic arguments below, $\lambda = \lambda_N$ determines the Wald-significance penalty-activation threshold of [Eq. \(S3\)](#): it may be deterministic or data-adaptive, provided it is measurable with respect to the outer training data and satisfies [Assumption S4\(b\)–\(c\)](#) (i.e. $1/\lambda_N \rightarrow \infty$ while $1/\lambda_N = o(\sqrt{N_{\text{all}}} b_{0,N})$) with probability tending to one; it is *not* a penalty parameter driven to infinity. Under fixed separation this condition is the simpler $\lambda_N \rightarrow 0$ and $\lambda_N \sqrt{N_{\text{all}}} \rightarrow \infty$. The inner validation construction preserves the cross-fitting independence needed for the influence-function expansion because the training summaries used to compute $\hat{\eta}^{(-k,-m)}(\lambda)$, the validation summaries used to choose $\hat{\lambda}^{(-k)}$, and the final outer evaluation fold $\mathcal{D}_k$ are separated. Consequently neither $\hat{\lambda}^{(-k)}$ nor $\hat{\eta}^{(-k)}$ uses the outer evaluation fold $\mathcal{D}_k$. In the production implementation, the locked threshold $1/\lambda = 1$ activates the penalty when $\hat{t}_j^\tau > 1$. Crossing this threshold initiates soft shrinkage and is not interpreted as a stand-alone hypothesis test or as guaranteed source exclusion.

Remark S7 (Soft-thresholding interpretation). The penalty in [Eq. \(S3\)](#) is adaptive ℓ_1 . For a diagonal Hessian of $\hat{L}_\tau$, each coordinate minimizes

$$\widehat{H}_{\tau,jj} \eta_j^2 + \widehat{b}_{\tau,j} \eta_j + (\lambda \hat{t}_j^\tau - 1)_+ |\eta_j|, \quad \widehat{b}_{\tau,j} := \left. \frac{\partial \hat{L}_\tau}{\partial \eta_j} \right|_{\eta=0} = N_{\text{all}} \frac{2(\hat{C}_{ot,s_j}^\tau - \hat{V}_{ot}^\tau)}{N_t},$$

where $\widehat{b}_{\tau,j} = O_p(1)$ is the asymptotic-variance gradient at $\eta = 0$. Completing the square yields the explicit solution

$$\widehat{\eta}_j = \text{SoftThreshold}\left(\eta_j^{\text{OLS}}, \frac{(\lambda \widehat{t}_j^\tau - 1)_+}{2\widehat{H}_{\tau,jj}}\right), \quad \eta_j^{\text{OLS}} := -\frac{\widehat{b}_{\tau,j}}{2\widehat{H}_{\tau,jj}},$$

where η_j^{OLS} is the unpenalized variance-minimizing coordinate. Equivalently, the KKT selection rule is

$$\widehat{\eta}_j = 0 \iff |\widehat{b}_{\tau,j}| \leq (\lambda \widehat{t}_j^\tau - 1)_+,$$

with the analogous coupled KKT conditions for a non-diagonal Hessian.

S3.3 Informative Source Indices

For each source, write $\mu_{t,s_j}^{a,*} = \text{plim } \widehat{\mu}_{t,s_j}^a$. Define the TATE-compatible source set

$$\mathcal{J}_\tau^* = \{j : \mu_{t,s_j}^{1,*} - \mu_{t,s_j}^{0,*} = \tau\} = \{j : \text{plim } \widehat{\tau}_{t,s_j} = \tau\}.$$

Define the oracle-restricted space:

$$\mathbb{R}_{\mathcal{J}_\tau^*} := \{\eta \in \mathbb{R}^K : \eta_j = 0 \text{ for all } j \notin \mathcal{J}_\tau^*\}.$$

Let η_τ^* be the oracle TATE weights:

$$\eta_\tau^* = \arg \min_{\eta \in \mathbb{R}_{\mathcal{J}_\tau^*}} L_\tau^*(\eta), \quad L_\tau^*(\eta) = \text{Var}\{\sqrt{N_{\text{all}}}(\widehat{\tau}_{\text{agg}}(\eta) - \tau)\}.$$

Assumption S4 (Aggregation Regularity). Assume:

(a) (*Compatible sources and global curvature*) The set $\mathcal{J}_\tau^*$ is non-empty. The quadratic objective L_τ^* has a positive-definite Hessian on $\mathbb{R}^K$, with minimum eigenvalue bounded away from zero. Hence L_τ^* is coercive and has a finite unique minimizer on every coordinate subspace, including $\mathbb{R}_{\mathcal{J}_\tau^*}$.

(b) (*Separation / “ β -min” for biased sites*) Define

$$b_{0,N}^\tau := \min_{j \notin \mathcal{J}_\tau^*} |\text{plim}(\widehat{\tau}_{t,s_j}) - \tau|.$$

Assume $b_{0,N}^\tau \sqrt{N_{\text{all}}} \rightarrow \infty$ and that the asymptotic variance of each candidate TATE discrepancy $\sqrt{N_{\text{all}}} \{\hat{\tau}_{t,s_j} - \hat{\tau}_{ot}\}$ is bounded away from zero.

(c) (*Threshold rate*) The penalty-activation threshold satisfies

$$\lambda \rightarrow 0, \quad \lambda \sqrt{N_{\text{all}}} b_{0,N}^\tau \rightarrow \infty,$$

equivalently $1/\lambda = o(\sqrt{N_{\text{all}}} b_{0,N}^\tau)$ while $1/\lambda \rightarrow \infty$.

(d) (*Fixed source set*) The number of source sites K is fixed.

(e) (*Compatible-source contrast expansion*) For every $j \in \mathcal{J}_\tau^*$, the centered contrast satisfies

$$\sqrt{N_{t,s_j}} [\{\hat{\mu}_{t,s_j}^1 - \mu_{t,s_j}^{1,*}\} - \{\hat{\mu}_{t,s_j}^0 - \mu_{t,s_j}^{0,*}\}] = \frac{1}{\sqrt{N_{t,s_j}}} \sum_i \varphi_{t,s_j,i}^\tau + o_p(1),$$

with the corresponding multisite scaling, a centered finite-variance influence function, and the uniform foldwise version used by the cross-fitted estimator. Separate application of [Theorems S1](#) and [S2](#) is sufficient. If compatibility instead arises only because the two arm-specific pseudo-limit biases cancel, this contrast expansion is an additional assumption.

If λ is selected over a pre-specified sensitivity grid, the separation and threshold-window conditions are interpreted as high-probability conditions on the selected sequence. The primary implementation instead fixes $1/\lambda = 1$ before examining the outer evaluation fold. The oracle support recovery and single normal limit stated below therefore concern the rate-indexed sequence in [Assumption S4\(c\)](#); they are not asserted for the fixed-cutoff implementation without an additional fixed-threshold argument.

Remark S8 (When separation fails / local alternatives). [Assumption S4\(b\)](#) is used only for support recovery. Under local TATE alternatives $b_{0,N}^\tau = O(N_{\text{all}}^{-1/2})$, exact selection can fail because $\hat{t}_j^\tau$ remains $O_p(1)$. The added first-order bias is controlled by

$$\left| \sum_{j \notin \mathcal{J}_\tau^*} \eta_j \{\text{plim}(\hat{\tau}_{t,s_j}) - \tau\} \right|,$$

which is negligible when the included TATE deviations are $o(N_{\text{all}}^{-1/2})$.

S3.4 Connection to the Original FACE Squared-Discrepancy Objective

The main FACE-HD estimator uses the truncated-Wald objective in Eq. (S3). This subsection records the direct analogue of the original FACE adaptive-aggregation objective (Han et al. 2025, Equation (11)), which uses the raw squared discrepancy rather than the studentized Wald statistic. With the notation of Section S3.3, define

$$\hat{\eta}_{\text{sq}} = \arg \min_{\eta \in \mathbb{R}^K} \left\{ \hat{L}_\tau(\eta) + \lambda_{\text{sq},N} \sum_{j=1}^K |\eta_j| (\text{disc}_j^\tau)^2 \right\}, \quad \text{disc}_j^\tau = \hat{\tau}_{ot} - \hat{\tau}_{t,s_j}, \quad (\text{S4})$$

where $\hat{L}_\tau(\eta)$ is the same rescaled TATE-variance objective used in Eq. (S3).

The oracle argument is the same as the proof of Lemma S11, but the rate window is different. Suppose Assumptions S1 to S3 hold and that the global-curvature and fixed- K parts of Assumption S4(a),(d) hold. If the raw-discrepancy tuning parameter satisfies

$$\lambda_{\text{sq},N} \rightarrow \infty, \quad \lambda_{\text{sq},N} = o(N_{\text{all}}^{1/2}), \quad \lambda_{\text{sq},N} (b_{0,N}^\tau)^2 \rightarrow \infty, \quad (\text{S5})$$

then the squared-discrepancy objective has the same oracle-selection behavior:

$$P(\hat{\eta}_{\text{sq},j} = 0 \text{ for all } j \notin \mathcal{J}_\tau^*) \rightarrow 1, \quad \|\hat{\eta}_{\text{sq},\mathcal{J}_\tau^*} - \eta_{\tau,\mathcal{J}_\tau^*}^*\|_2 = o_p(1).$$

For each $j \in \mathcal{J}_\tau^*$, the two arm expansions imply $\text{disc}_j^\tau = O_p(N_{\text{all}}^{-1/2})$, hence

$$\lambda_{\text{sq},N} (\text{disc}_j^\tau)^2 = O_p(\lambda_{\text{sq},N}/N_{\text{all}}) = o_p(N_{\text{all}}^{-1/2}),$$

so the penalty is negligible on the oracle subspace. For each $j \notin \mathcal{J}_\tau^*$, $|\text{disc}_j^\tau| \geq b_{0,N}^\tau/2$ w.p.a.1; therefore

$$\lambda_{\text{sq},N} (\text{disc}_j^\tau)^2 \geq \frac{1}{4} \lambda_{\text{sq},N} (b_{0,N}^\tau)^2 \rightarrow \infty \quad \text{w.p.a.1.}$$

As in Lemma S11, coercivity first gives $\|\hat{\eta}_{\text{sq}}\|_2 = O_p(1)$, and the gradient of the quadratic variance part is therefore $O_p(1)$ along the segment joining the optimizer to its coordinate-zero

competitor. The diverging adaptive penalty makes that competitor strictly favorable unless $\hat{\eta}_{\text{sq},j} = 0$, forcing zero weights for biased sources.

Under fixed TATE separation, this reduces to the original FACE scaling $\lambda_{\text{sq},N} \rightarrow \infty$ and $\lambda_{\text{sq},N} = o(N_{\text{all}}^{1/2})$. The truncated-Wald objective needs only the weaker studentized TATE separation in [Assumption S4](#).

S3.5 Aggregation Main Results

For later use in the aggregation proofs, we restate the pairwise input result that is repeatedly invoked below.

Theorem (restated). *Suppose [Assumptions S1](#) to [S3](#) hold. Assume ϑ^* is the unique population target pair induced by the calibrated losses under the alignment condition in [Assumption S3\(c\)](#), implying [Lemma S5](#) holds. If further, at least one nuisance model (outcome or weighting) is correctly specified in the sense of [Lemma S1](#) (so $\mu^1(\vartheta^*) = \mu^1$), the calibrated nuisance sparsities satisfy*

$$\frac{(s_\alpha + s_\gamma)^2 \log^2(p)}{N} \longrightarrow 0,$$

and $V_{t,s}^ > 0$, then the cross-fitted estimator is asymptotically normal. Here s_α and s_γ denote the calibrated sparsities from [Definition S1](#):*

$$\sqrt{N} \left(\hat{\mu}_{t,s}^{1,\text{CF}} - \mu^1 \right) \xrightarrow{d} \mathcal{N}(0, V_{t,s}^*).$$

If the initial nuisance estimators enter the two-level algorithm separately from the calibrated update, their stochastic effect is absorbed through [Assumption S3\(c\)–\(d\)](#). The finite-sample KKT perturbation from the ℓ_1 penalties is likewise absorbed in these nuisance-rate conditions and is $o_p(N^{-1/2})$ under the stated sparsity scaling.

Theorem S3 (Asymptotic Normality of Direct-TATE FACE-HD). *Suppose the pairwise assumptions hold and [Assumption S4](#) holds. For $j \in \mathcal{J}_\tau^*$, let $N_{t,s_j} = N_t + N_{s_j}$ and let $s_{\text{tot},j}$*

bound the combined arm-specific nuisance sparsity. If

$$\max_{j \in \mathcal{J}_\tau^*} \frac{s_{\text{tot},j}^2 \log^2(p)}{N_{t,s_j}} \rightarrow 0,$$

and $L_\tau^*(\eta_\tau^*) > 0$, then

$$\sqrt{N_{\text{all}}}(\hat{\tau}_{\text{agg}} - \tau) \xrightarrow{d} \mathcal{N}\{0, L_\tau^*(\eta_\tau^*)\}.$$

Moreover, the within-site-centered pseudo-value estimator in [Eq. \(11\)](#) is consistent:

$$\hat{V}_\tau \xrightarrow{p} L_\tau^*(\eta_\tau^*),$$

while, uniformly over the fixed outer folds, $\hat{L}_{\tau,-k}\{\hat{\eta}^{(-k)}\} \xrightarrow{p} L_\tau^*(\eta_\tau^*)$; these two estimators need not be numerically identical in finite samples. Furthermore,

$$\max_{k \leq K_f} \|\hat{\eta}^{(-k)} - \eta_\tau^*\|_2 = o_p(1), \quad P\left(\hat{\eta}_j^{(-k)} = 0 \text{ for every } j \notin \mathcal{J}_\tau^*, k \leq K_f\right) \rightarrow 1,$$

and

$$\sqrt{\frac{N_{\text{all}}}{\hat{V}_\tau}}(\hat{\tau}_{\text{agg}} - \tau) \xrightarrow{d} \mathcal{N}(0, 1).$$

Proposition S1 (TATE Efficiency Gain). *Under the conditions of [Theorem S3](#), direct-TATE FACE-HD is asymptotically at least as efficient as target-only TATE:*

$$L_\tau^*(\eta_\tau^*) \leq L_\tau^*(0) = V_{\tau, \text{target}},$$

with strict inequality whenever there exists $j \in \mathcal{J}_\tau^*$ such that

$$\text{Var}(\varphi_{t,s_j}^\tau - \varphi_{ot}^\tau) > 0 \quad \text{and} \quad \text{Cov}(\varphi_{ot}^\tau, \varphi_{t,s_j}^\tau - \varphi_{ot}^\tau) \neq 0.$$

S3.6 Aggregation Supporting Lemmas

To keep the remaining proofs readable, use the generic shorthand

$$\theta = \tau, \quad \hat{\theta}_{ot} = \hat{\tau}_{ot}, \quad \hat{\theta}_{t,s_j} = \hat{\tau}_{t,s_j}, \quad \hat{\theta}_{\text{agg}} = \hat{\tau}_{\text{agg}}, \quad \mathcal{J}^* = \mathcal{J}_\tau^*, \quad \eta^* = \eta_\tau^*, \quad L^* = L_\tau^*,$$

and suppress the superscript τ on the treatment-contrast influence components. Thus every object in the supporting lemmas below is TATE-level.

Lemma S10 (Influence Function Representation). Under [Assumptions S1 to S3](#) and [Assumption S4\(a\),\(e\)](#), the aggregated estimator admits the asymptotic linear representation:

$$\sqrt{N_{\text{all}}} \{\hat{\theta}_{\text{agg}}(\eta^*) - \theta\} = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1)$$

where $\varphi_{\text{agg},i}(\eta^*)$ is the aggregated influence function:

$$\varphi_{\text{agg},i}(\eta^*) = \left(1 - \sum_{j \in \mathcal{J}^*} \eta_j^*\right) \varphi_{ot,i} + \sum_{j \in \mathcal{J}^*} \eta_j^* \varphi_{t,s_j,i},$$

with $\varphi_{t,s_j,i}$ denoting the (rescaled) IF contribution of source-assisted estimator j under $\sqrt{N_{\text{all}}}$ scaling. **Convention:** Each $\varphi_{ot,i}$ (resp. $\varphi_{t,s_j,i}$) is understood as zero for observations not in the target data $\mathcal{D}_t$ (resp. not in $\mathcal{D}_t \cup \mathcal{D}_{s_j}$), and includes the constant rescaling needed to convert its natural $\sqrt{N_t}$ (resp. $\sqrt{N_{t,s_j}}$) linear representation into the common $\sqrt{N_{\text{all}}}$ scaling.

Proof. By the compatible-source contrast expansion in [Assumption S4\(e\)](#)—which follows by applying the pairwise expansion to both arms when they are separately identified—for every $j \in \mathcal{J}^*$,

$$\sqrt{N_{t,s_j}} \{\hat{\theta}_{t,s_j} - \text{plim}(\hat{\theta}_{t,s_j})\} = \frac{1}{\sqrt{N_{t,s_j}}} \sum_{i=1}^{N_{\text{all}}} \bar{\varphi}_{t,s_j,i} + o_p(1),$$

where $\bar{\varphi}_{t,s_j,i}$ is understood as zero for observations not in $\mathcal{D}_t \cup \mathcal{D}_{s_j}$. Since $N_{t,s_j}/N_{\text{all}} \rightarrow c_j \in (0, 1)$, the above is equivalent to

$$\sqrt{N_{\text{all}}} \{\hat{\theta}_{t,s_j} - \text{plim}(\hat{\theta}_{t,s_j})\} = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{t,s_j,i} + o_p(1)$$

for a rescaled IF $\varphi_{t,s_j,i}$. The target-only estimator has the analogous form:

$$\sqrt{N_{\text{all}}} (\hat{\theta}_{ot} - \theta) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{ot,i} + o_p(1).$$

For $j \in \mathcal{J}^*$, $\text{plim}(\hat{\theta}_{t,s_j}) = \theta$.

By linearity, the aggregated estimator with fixed $\eta = \eta^*$ satisfies:

$$\sqrt{N_{\text{all}}} \{\hat{\theta}_{\text{agg}}(\eta^*) - \theta\} = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1).$$

The remainder remains $o_p(1)$ because K is fixed. □

Lemma S11 (Selection Consistency). Under [Assumptions S1 to S3](#) and [Assumption S4](#), the penalized estimator $\hat{\eta}$ from [Eq. \(S3\)](#) satisfies:

1. (**Oracle-subspace consistency**) $\|\hat{\eta} - \eta^*\|_2 = o_p(1)$
2. (**Selection**) $\hat{\eta}_j = 0$ for all $j \notin \mathcal{J}^*$ with probability approaching 1.

Proof. We give a self-contained argument; it parallels the FACE proof in [Han et al. \(2025\)](#) (Lemma S5) but the objective here is an explicit quadratic form. Recall $\hat{L}(\eta) = \eta^T \hat{H} \eta + \hat{\mathbf{d}}^T \eta + \hat{c}$ and its population analogue $L^*(\eta) = \eta^T H \eta + \mathbf{d}^T \eta + c$.

Part 1 (Consistency on the oracle subspace). Define the oracle-restricted estimator

$$\tilde{\eta} = \arg \min_{\eta \in \mathbb{R}_{\mathcal{J}^*}} \left\{ \hat{L}(\eta) + \sum_{j=1}^K (\lambda \hat{t}_j^\tau - 1)_+ |\eta_j| \right\}.$$

For $j \in \mathcal{J}^*$, $\text{disc}_j^\tau = \hat{\theta}_{ot} - \hat{\theta}_{t,s_j} = O_p(N_{\text{all}}^{-1/2})$ and $\hat{t}_j^\tau = O_p(1)$. Under $1/\lambda \rightarrow \infty$, the penalty is zero on every oracle coordinate w.p.a.1, so $\tilde{\eta}$ coincides with the unpenalized TATE-variance minimizer on $\mathbb{R}_{\mathcal{J}^*}$.

Next, by [Lemma S9](#), the quadratic coefficients satisfy $\|\hat{H} - H\|_{\text{op}} = o_p(1)$ and $\|\hat{\mathbf{d}} - \mathbf{d}\|_2 = o_p(1)$. Let $H_{\mathcal{J}^*}$ and $\hat{H}_{\mathcal{J}^*}$ denote the principal submatrices corresponding to $\mathcal{J}^*$, and similarly let $\mathbf{d}_{\mathcal{J}^*}$ and $\hat{\mathbf{d}}_{\mathcal{J}^*}$ denote the corresponding subvectors. Because K is fixed, $H_{\mathcal{J}^*}$ is a fixed-dimensional positive definite matrix by the global-curvature condition, hence $\|H_{\mathcal{J}^*}^{-1}\|_{\text{op}} < \infty$ and $\|\hat{H}_{\mathcal{J}^*}^{-1}\|_{\text{op}} = O_p(1)$. More explicitly, by Weyl's inequality and [Lemma S9](#), $\lambda_{\min}(\hat{H}_{\mathcal{J}^*}) \geq \lambda_{\min}(H_{\mathcal{J}^*}) - \|\hat{H}_{\mathcal{J}^*} - H_{\mathcal{J}^*}\|_{\text{op}}$, which is bounded away from 0 w.p.a.1, so $\hat{H}_{\mathcal{J}^*}$ is invertible and $\|\hat{H}_{\mathcal{J}^*}^{-1}\|_{\text{op}} = O_p(1)$. Because the penalty is zero on the oracle subspace w.p.a.1 and both quadratic objectives are coercive and strictly convex there, their unconstrained minimizers satisfy the first-order conditions $2\hat{H}_{\mathcal{J}^*}\eta + \hat{\mathbf{d}}_{\mathcal{J}^*} = 0$ and $2H_{\mathcal{J}^*}\eta + \mathbf{d}_{\mathcal{J}^*} = 0$, so

$$\eta_{\mathcal{J}^*}^* = -\frac{1}{2}H_{\mathcal{J}^*}^{-1}\mathbf{d}_{\mathcal{J}^*}, \quad \tilde{\eta}_{\mathcal{J}^*} = -\frac{1}{2}\hat{H}_{\mathcal{J}^*}^{-1}\hat{\mathbf{d}}_{\mathcal{J}^*} + o_p(1).$$

To make the perturbation step explicit, write

$$\tilde{\eta}_{\mathcal{J}^*} - \eta_{\mathcal{J}^*}^* = -\frac{1}{2} \left(\widehat{H}_{\mathcal{J}^*}^{-1} \widehat{\mathbf{d}}_{\mathcal{J}^*} - H_{\mathcal{J}^*}^{-1} \mathbf{d}_{\mathcal{J}^*} \right) + o_p(1).$$

Then

$$\widehat{H}_{\mathcal{J}^*}^{-1} \widehat{\mathbf{d}}_{\mathcal{J}^*} - H_{\mathcal{J}^*}^{-1} \mathbf{d}_{\mathcal{J}^*} = \widehat{H}_{\mathcal{J}^*}^{-1} (\widehat{\mathbf{d}}_{\mathcal{J}^*} - \mathbf{d}_{\mathcal{J}^*}) + (\widehat{H}_{\mathcal{J}^*}^{-1} - H_{\mathcal{J}^*}^{-1}) \mathbf{d}_{\mathcal{J}^*}.$$

The first term is $o_p(1)$ since $\|\widehat{H}_{\mathcal{J}^*}^{-1}\|_{\text{op}} = O_p(1)$ and $\|\widehat{\mathbf{d}} - \mathbf{d}\|_2 = o_p(1)$. For the second term, use $\widehat{H}_{\mathcal{J}^*}^{-1} - H_{\mathcal{J}^*}^{-1} = \widehat{H}_{\mathcal{J}^*}^{-1} (H_{\mathcal{J}^*} - \widehat{H}_{\mathcal{J}^*}) H_{\mathcal{J}^*}^{-1}$, so

$$\|\widehat{H}_{\mathcal{J}^*}^{-1} - H_{\mathcal{J}^*}^{-1}\|_{\text{op}} \leq \|\widehat{H}_{\mathcal{J}^*}^{-1}\|_{\text{op}} \|\widehat{H} - H\|_{\text{op}} \|H_{\mathcal{J}^*}^{-1}\|_{\text{op}} = o_p(1),$$

and hence the second term is also $o_p(1)$. Therefore $\|\tilde{\eta} - \eta^*\|_2 = o_p(1)$. Finally, since $\hat{\eta}$ agrees with $\tilde{\eta}$ w.p.a.1 by Part 2 below, the same consistency holds for $\hat{\eta}$.

Part 2 (Selection of biased sites). First, the unconstrained penalized minimizer is stochastically bounded. By [Assumption S4\(a\)](#), coefficient consistency, and Weyl's inequality, there is a constant $h_0 > 0$ such that $\lambda_{\min}(\widehat{H}) \geq h_0$ w.p.a.1. Since the penalty is nonnegative and the objective at $\hat{\eta}$ is no larger than its value at zero,

$$h_0 \|\hat{\eta}\|_2^2 - \|\widehat{\mathbf{d}}\|_2 \|\hat{\eta}\|_2 \leq 0.$$

Because $\|\widehat{\mathbf{d}}\|_2 = O_p(1)$, this implies $\|\hat{\eta}\|_2 = O_p(1)$. Hence the quadratic gradient is $O_p(1)$ uniformly along any line segment joining $\hat{\eta}$ to a vector formed by setting one of its coordinates to zero:

$$\sup_{\bar{\eta} \text{ on such a segment}} \left| \frac{\partial \widehat{L}}{\partial \eta_j}(\bar{\eta}) \right| \leq 2 \|\widehat{H}\|_{\text{op}} \|\hat{\eta}\|_2 + \|\widehat{\mathbf{d}}\|_2 = O_p(1).$$

For $j \notin \mathcal{J}^*$, let $b_{j,N}^\tau = |\text{plim}(\hat{\theta}_{t,s_j}) - \theta|$. Then

$$\hat{t}_j^\tau = \frac{|\text{disc}_j^\tau|}{\widehat{\text{SE}}(\text{disc}_j^\tau)} = \Theta_p(\sqrt{N_{\text{all}} b_{j,N}^\tau}) \rightarrow \infty,$$

and $(\lambda \hat{t}_j^\tau - 1)_+ \rightarrow \infty$ by [Assumption S4\(b\)–\(c\)](#).

Let $\hat{\eta}^{(j=0)}$ equal $\hat{\eta}$ with coordinate j set to zero. This is a feasible point in $\mathbb{R}^K$. The mean-value theorem and the preceding gradient bound give

$$\hat{L}\{\hat{\eta}^{(j=0)}\} - \hat{L}(\hat{\eta}) \leq \sup_{\bar{\eta} \text{ on the segment}} \left| \frac{\partial \hat{L}}{\partial \eta_j}(\bar{\eta}) \right| |\hat{\eta}_j| = O_p(1) |\hat{\eta}_j|.$$

The penalty term decreases by $(\lambda \hat{t}_j^\tau - 1)_+ |\hat{\eta}_j|$ when $\hat{\eta}_j$ is set to zero. Since this coefficient diverges while the quadratic-gradient bound is $O_p(1)$, every incompatible coordinate is zero w.p.a.1. Therefore $\hat{\eta} \in \mathbb{R}_{\mathcal{J}^*}$ w.p.a.1, hence $\hat{\eta} = \tilde{\eta}$ w.p.a.1. $\square$

Lemma S12 (Asymptotic Equivalence). *Under [Assumptions S1](#) to [S3](#) and [Assumption S4](#):*

$$\sqrt{N_{\text{all}}} \{\hat{\theta}_{\text{agg}}(\hat{\eta}) - \hat{\theta}_{\text{agg}}(\eta^*)\} = o_p(1)$$

Proof. The difference is:

$$\sqrt{N_{\text{all}}} \{\hat{\theta}_{\text{agg}}(\hat{\eta}) - \hat{\theta}_{\text{agg}}(\eta^*)\} = \sum_{j=1}^K (\hat{\eta}_j - \eta_j^*) \sqrt{N_{\text{all}}} (\hat{\theta}_{t,s_j} - \hat{\theta}_{ot})$$

For $j \in \mathcal{J}^*$, weight consistency and the two arm-specific pairwise expansions give $|\hat{\eta}_j - \eta_j^*| = o_p(1)$ and $\sqrt{N_{\text{all}}} (\hat{\theta}_{t,s_j} - \hat{\theta}_{ot}) = O_p(1)$.

For $j \notin \mathcal{J}^*$: By [Lemma S11](#)(2), $\hat{\eta}_j = 0$ w.p.a.1, and $\eta_j^* = 0$ by definition (oracle excludes biased sites). Thus the difference is zero w.p.a.1. Because K is fixed, summing over j yields $o_p(1)$. $\square$

Remark S9 (Growing number of sources K). The fixed- K argument extends only with additional concentration. A sufficient condition for the plug-in term is

$$\|\hat{\eta} - \eta^*\|_1 \max_{j \leq K} \left| \sqrt{N_{\text{all}}} (\hat{\theta}_{t,s_j} - \hat{\theta}_{ot}) \right| = o_p(1),$$

which typically requires both a sparsity assumption on η^* and concentration bounds scaling with $\log K$.

S3.7 Proofs of Aggregation Results

Proof of Theorem S3. Combining Lemma S10, Lemma S11, and Lemma S12:

$$\sqrt{N_{\text{all}}}(\hat{\theta}_{\text{agg}} - \theta) = \frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) + o_p(1).$$

As in Lemma S3, the observations are independent with a multi-site structure and site proportions converging to constants. The aggregated influence function contributions $\{\varphi_{\text{agg},i}(\eta^*)\}_{i=1}^{N_{\text{all}}}$ are therefore independent with mean zero and finite variance $L^*(\eta^*)$. Under Assumption S2(b), these contributions have a uniform $(2+\delta)$ moment, so the same Lyapunov argument as in Lemma S3 applies and yields:

$$\frac{1}{\sqrt{N_{\text{all}}}} \sum_{i=1}^{N_{\text{all}}} \varphi_{\text{agg},i}(\eta^*) \xrightarrow{d} \mathcal{N}(0, L^*(\eta^*)).$$

For variance estimation consistency, first fix an outer fold. By Lemma S11(1), $\|\hat{\eta}^{(-k)} - \eta^*\|_2 = o_p(1)$. By the uniform convergence of $\hat{L}_{\tau,-k}$ to L^* and local Lipschitz continuity of the quadratic map $L^*(\eta) = \eta^T H \eta + \mathbf{d}^T \eta + c$ in a neighborhood of η^* (e.g., on $\{\eta : \|\eta - \eta^*\|_2 \leq R\}$, $|L^*(\eta_1) - L^*(\eta_2)| \leq (2\|H\|_{\text{op}}R + \|\mathbf{d}\|_2)\|\eta_1 - \eta_2\|_2$):

$$|\hat{L}_{\tau,-k}\{\hat{\eta}^{(-k)}\} - L^*(\eta^*)| = o_p(1).$$

This holds uniformly over $k \leq K_f$ because the number of folds is fixed. For the distinct held-out variance estimator, evaluate each observation's pseudo-value with its fold-specific weight. Weight consistency, the local L_2 stability of the cross-fitted influence components, and the within-site conditional law of large numbers imply that its within-site first and second moments differ by $o_p(1)$ from those of the fixed oracle-weight pseudo-values. Therefore the WSS construction in Remark S2 gives $\hat{V}_\tau \xrightarrow{p} L^*(\eta^*)$.

Finally, write

$$\sqrt{\frac{N_{\text{all}}}{\hat{V}_\tau}}(\hat{\theta}_{\text{agg}} - \theta) = \left\{ \sqrt{\frac{L^*(\eta^*)}{\hat{V}_\tau}} \right\} \left\{ \frac{\sqrt{N_{\text{all}}}(\hat{\theta}_{\text{agg}} - \theta)}{\sqrt{L^*(\eta^*)}} \right\}.$$

The second factor converges to $\mathcal{N}(0, 1)$ and $\widehat{V}_\tau \xrightarrow{p} L^*(\eta^*)$, so the prefactor converges to one. Therefore the product converges in distribution to $\mathcal{N}(0, 1)$ by Slutsky’s theorem. $\square$

Proof of Proposition S1. Since η^* minimizes the TATE variance and $\eta = 0$ is feasible, $L^*(\eta^*) \leq L^*(0) = V_{\tau, \text{target}}$.

For strict improvement, consider one informative source $j \in \mathcal{J}^*$. The optimal projection is:

$$\eta_j^* = -\frac{\text{Cov}(\varphi_{ot}^\tau, \varphi_{t,s_j}^\tau - \varphi_{ot}^\tau)}{\text{Var}(\varphi_{t,s_j}^\tau - \varphi_{ot}^\tau)}.$$

If the covariance is non-zero and the denominator is positive, the minimizing weight is non-zero and strictly decreases variance relative to $\eta = 0$, so $L^*(\eta^*) < L^*(0)$. $\square$

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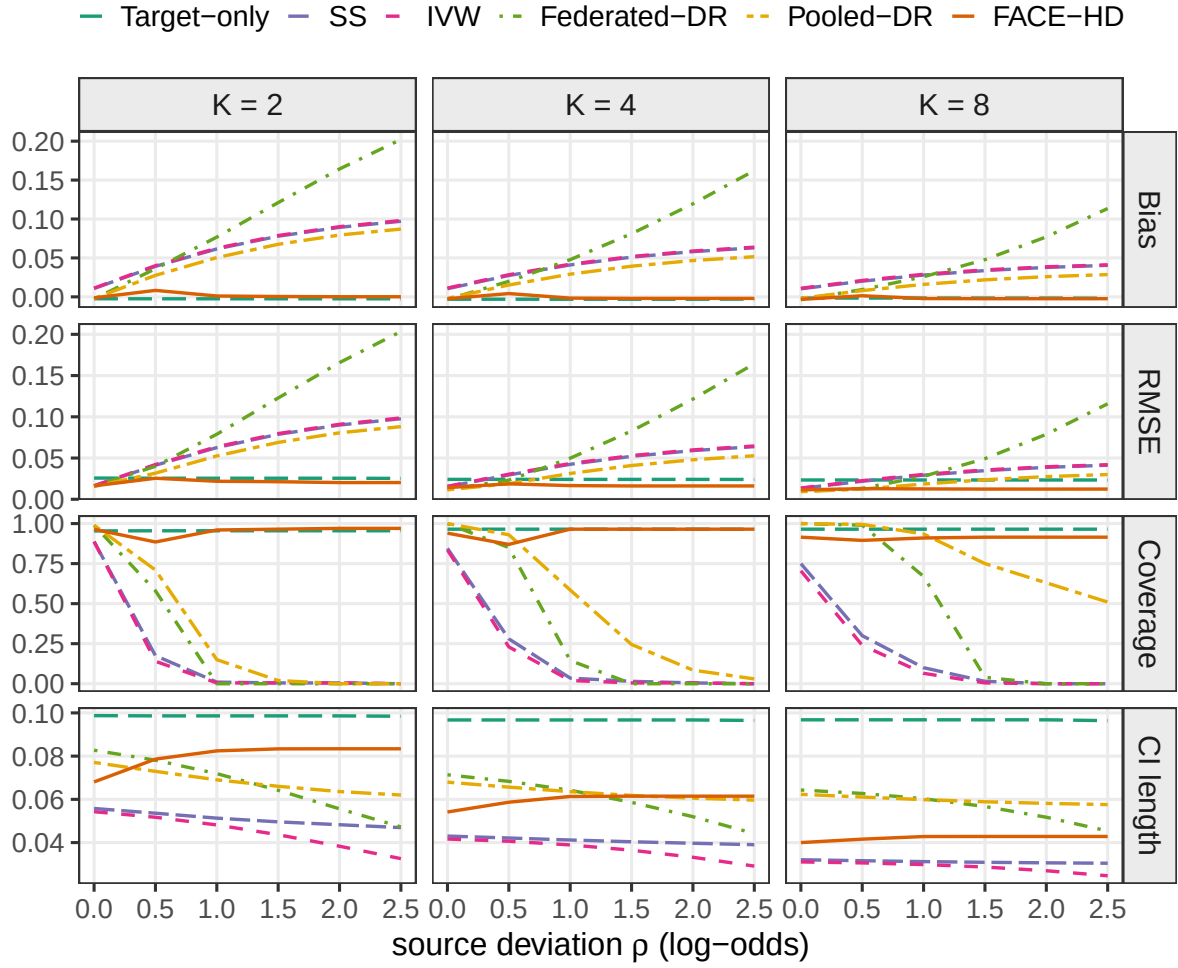


Figure S1: Legacy treated-mean negative-transfer simulation, $C(2)$, $p = 50$. Panels show bias, RMSE, 95% coverage, and CI length for μ^1 as the source deviation ρ increases, with columns for $K = 2, 4, 8$ source sites. FACE-HD remains stable while the naive aggregators lose coverage.

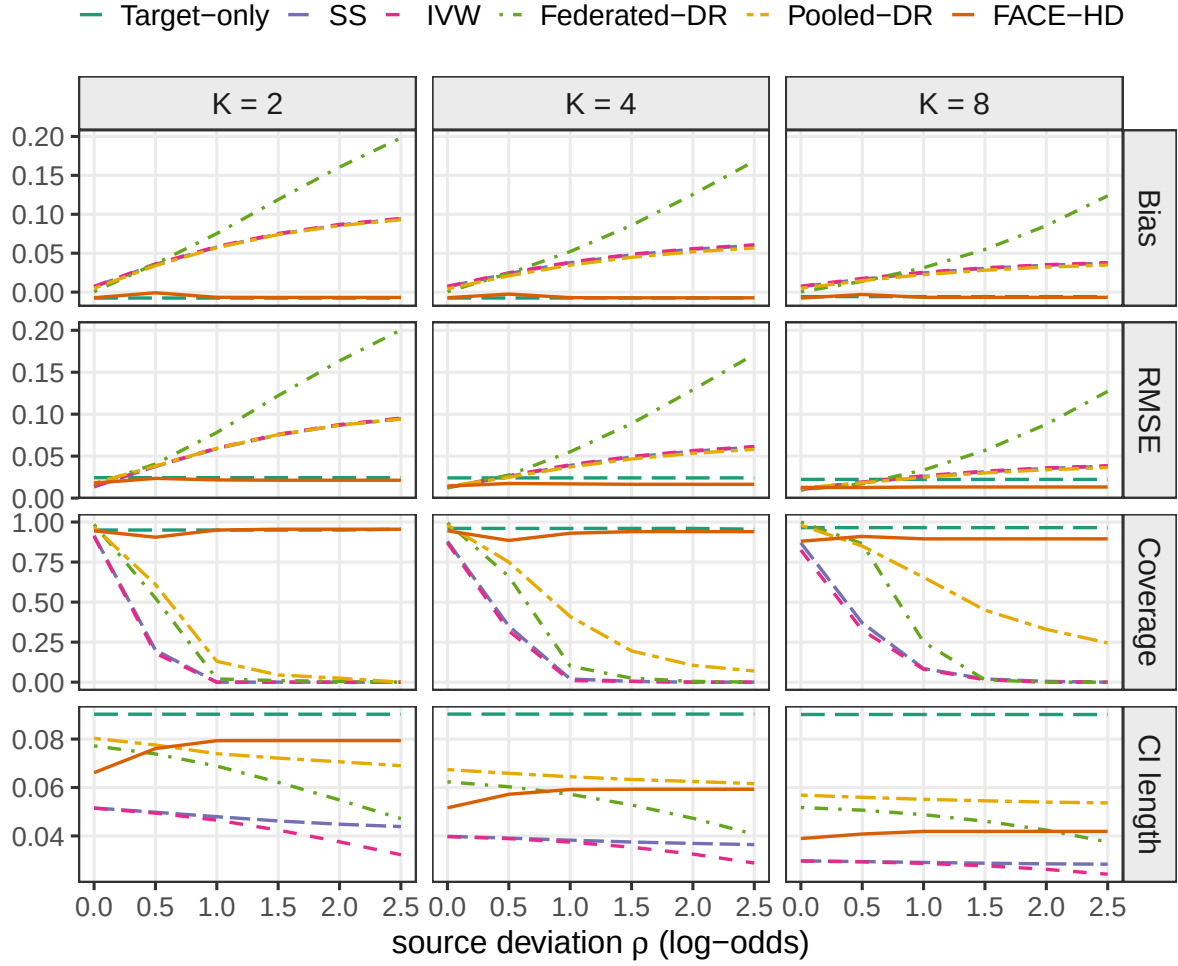


Figure S2: Legacy treated-mean negative-transfer simulation, $C(3)$, $p = 50$. Same layout as [figure S1](#); the site/propensity model is misspecified.

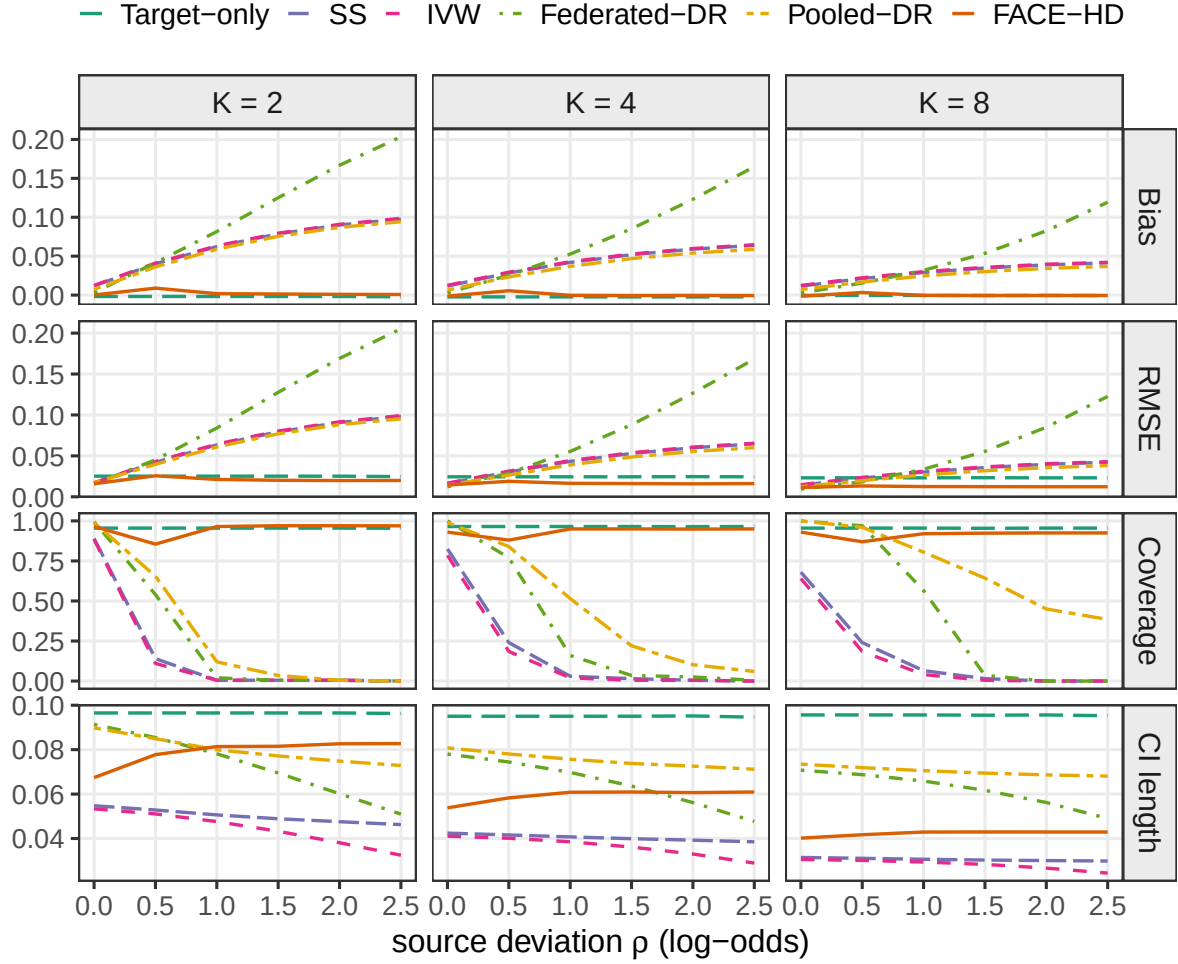


Figure S3: Legacy treated-mean negative-transfer simulation, $C(1)$, $p = 50$. Same layout as figure S1; the conclusions match the $p = 100$ case.

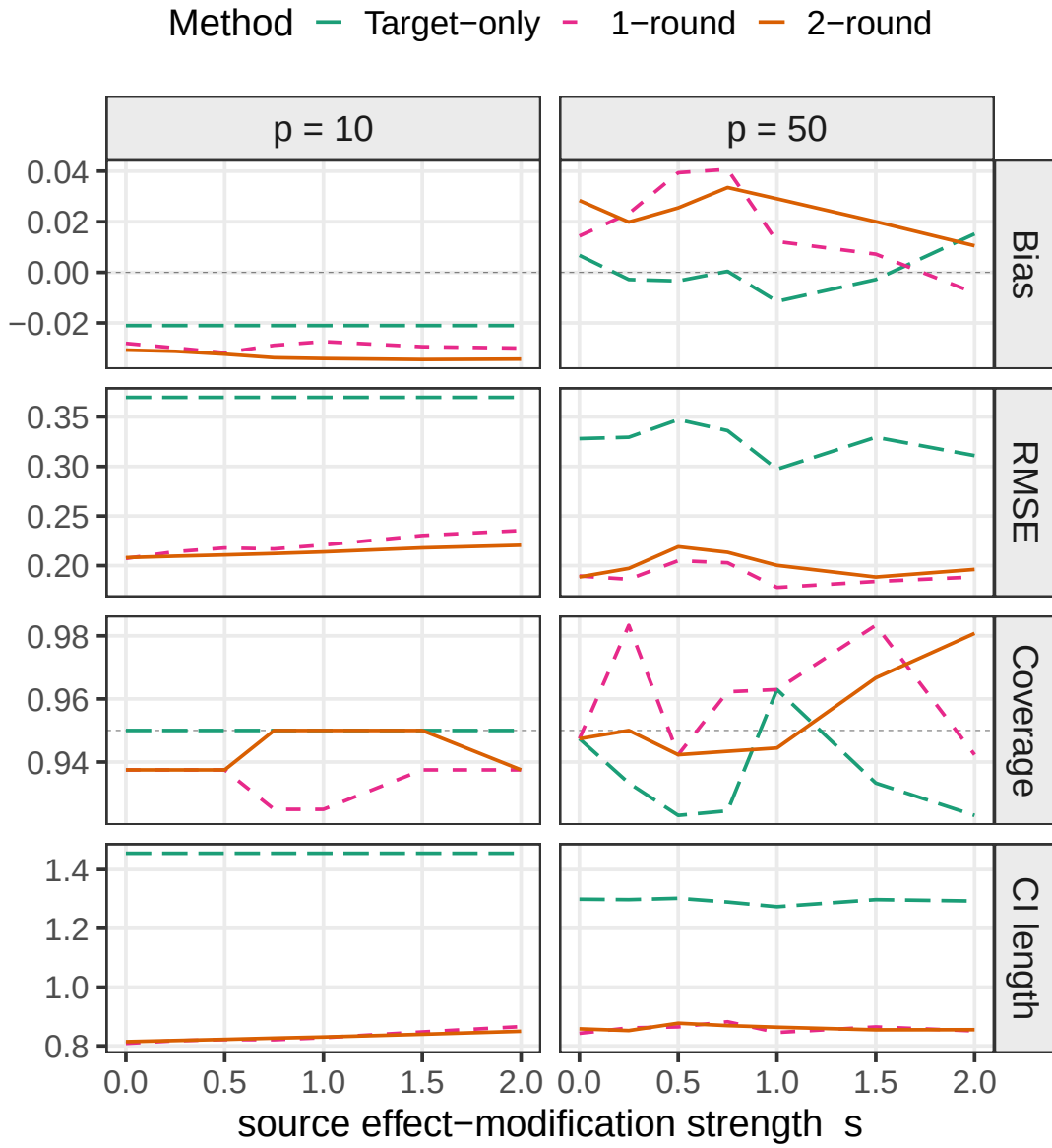


Figure S4: Archived one-round versus two-round arm-wise TATE FACE-HD for the continuous-outcome experiment. Panels show TATE bias, RMSE, 95% coverage, and CI length as the effect-modification strength s increases, for $p = 10$ and $p = 50$. Cells contain 80 completed replicates at $p = 10$ and 52–60 at $p = 50$; the comparison is descriptive, and the two variants track closely.